PurePy

Language Specification, v0.16.0

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Summary

This chapter collects the notation of the specification: the metavariables (Figure S1), the judgement forms with the figure defining each (Figure S2), and the auxiliary functions with the section defining each (Figure S3).

Names and literals		
x, f, c	variable; idiomatically f a function and c a class	§2.2
q	qualified name of a module or class	Figure 2.2
n	number, never negative; also sequence length and exit status	Figure 2.3
w	string	Figure 2.3
ℓ	literal	Figure 2.3
$\odot$, $\ominus$	binary and unary operator	Figure 2.5
Syntax		
e	expression	Figure 2.3
p	pattern	Figure 2.3
g	comprehension qualifier	Figure 2.3
b	case of a `match`	Figure 2.2
s	statement	Figure 2.2
d	function definition	Figure 2.2
τ, σ	type, a type expression whose class names are all marked	Figure 2.4
ψ	type expression, a type as written	Figure 2.4
v	primitive type	Figure 2.4
ι	import	Figure 2.2
ϕ	field list of a dataclass	Figure 2.2
t	top-level statement	Figure 2.2
μ	module body	Figure 2.2
u	argument: an expression, pattern or value	Chapter 2
a	argument list	Figure 2.3
Programs		
$\mathcal{M}$	program, a finite map from module names to bodies	§2.4
$\mathcal{P}$	predefined modules, each with its environment and context	§2.4
Contexts		
Γ, Δ	context, a finite map from variables to context entries	Definition 2.1
θ	context entry	Definition 2.1
C, D	class, a fully qualified class name; also the type of its instances	Definition 2.2
Σ	class table, a finite map from classes to class entries	Definition 2.3
R	static outcome of a statement	§3.3
ζ	optional return type, τ or ε	Figure 3.9
Shapes		
k, l	shape, denoting values of a type, idiomatically l residual	Figure 3.12
H	finite set of heads excluded by a shape	Figure 3.12
h	head, a literal, class or list of length n	Figure 3.12
β	bindings of a dictionary shape, a finite map from keys to shapes	Figure 3.12
K, L	ordered sets of shapes, idiomatically K matched and L residual	Figure 3.13
Values and results		
v	value	Figure 4.1
ρ	environment, a finite map from variables to values	Figure 4.1
δ	dictionary entries	Figure 4.1
r	result of statement evaluation	Figure 4.1
o	outcome of statement evaluation	Figure 4.1
r	result of expression evaluation	Figure 4.1
m	match result	Figure 4.1
κ	termination kind	Figure 4.1
f	meaning of a primitive, a partial map from a sequence of values to a result	Figure 4.1

Figure S1: Metavariables used throughout specification

Judgement	Reading	Defined in
Well-formedness		
$\Sigma \vdash h : \tau$	head h has type τ	Figure 3.12
$\Sigma \vdash k : \tau$	shape k has type τ	Figure 3.12
$\Sigma; \Gamma \vdash k, p \rightsquigarrow_{\mathsf{h}} K, L$	shape k splits at the head tested by p, giving K and L	Figure 3.14
$\Sigma; \Gamma \vdash k, p \rightsquigarrow K, L \dashv \Delta$	pattern p matches shapes K of k, leaving L and binding Δ	Figure 3.13
$\Gamma \vdash q : \theta$	qualified name q denotes context entry θ	Figure 3.1
$\Sigma; \Gamma \vdash e \rhd \tau$	expression e synthesises type τ	Figure 3.4
$\Sigma; \Gamma \vdash e \lhd \tau$	expression e checks against type τ	Figure 3.7
$\Sigma; \Gamma \vdash \vec{g} : \Delta$	comprehension qualifiers $\vec{g}$ are well-formed and bind Δ	Figure 3.7
$\Sigma; \Gamma; \zeta \vdash s : R$	statement s has static outcome R, with return type ζ	Figure 3.9
$\Sigma; \Gamma; \zeta; \sigma \vdash L, \vec{b} : S, L'$	cases $\vec{b}$ have static outcome S, transforming residual L to L'	Figure 3.10
$\Sigma; q; \Gamma \vdash t : R \dashv \Sigma'$	top-level statement t of module q has static outcome R, extending Σ to Σ'	Figure 3.11
$\Sigma \vdash \Gamma, x \ \mathsf{imports}\ \theta \dashv \Sigma'$	name x imported from module signature Γ has context entry θ	Figure 3.15
$\Sigma; q \vdash \vec{\imath} : \Gamma \dashv \Sigma'$	imports $\vec{\imath}$ of module q yield context Γ	Figure 3.16
$\Sigma \vdash q, \theta \ \mathsf{loads\text{-}as}\ \theta' \dashv \Sigma'$	importing module q into context entry θ gives context entry θ'	Figure 3.16
$\Sigma \vdash_{\mathsf{load}} q : \Gamma \dashv \Sigma'$	module q statically loads with signature Γ	Figure 3.17
$\vdash \mathcal{M}$	program $\mathcal{M}$ is well-formed	Figure 3.17
Evaluation		
$\vdash v : \theta$	value v has context entry θ	Figure 4.2
$\vdash \rho : \Gamma$	environment ρ has context Γ	Figure 4.2
$\rho, p, v \rightsquigarrow m$	pattern p matches value v with match result m	Figure 4.12
$\rho, s \Rightarrow o$	statement s evaluates to outcome o	Figure 4.3
$q, \rho, t \Rightarrow o$	top-level statement t of module q evaluates to outcome o	Figure 4.5
$\rho, e \Rightarrow r$	expression e evaluates to result r	Figure 4.6
$\rho, \vec{e} \Rightarrow r$	expressions $\vec{e}$ evaluate, left to right, to result r	Figure 4.8
$\rho, \vec{g} \Rightarrow r$	comprehension qualifiers $\vec{g}$ evaluate to result r	Figure 4.11
$\rho, x \ \mathsf{imports}\ r$	name x imported from module environment ρ has result r	Figure 4.13
$q, \vec{\imath} \Rightarrow r$	imports $\vec{\imath}$ of module q evaluate to result r	Figure 4.14
$q, v \ \mathsf{loads\text{-}as}\ v'$	importing module q into value v gives value v'	Figure 4.14
$q \Rightarrow_{\mathsf{load}} r$	module q loads to result r	Figure 4.15
$\mathcal{M} \Rightarrow \kappa$	program $\mathcal{M}$ evaluates to termination kind κ	Figure 4.15

Figure S2: Judgement forms

Function	Result	Meaning	Section
Syntactic analyses			
assigns(t)	set of variables	variables assigned in t, not descending into nested functions	§A.1
binds(p), binds($\vec{g}$)	set of variables	variables bound by pattern p, or by generators of qualifiers $\vec{g}$	§A.1
captures(t), captures($\vec{g}$)	set of variables	variables of enclosing scope captured by lambdas and nested functions in t, or in $\vec{g}$	§A.1
captures$_e$(e)	set of variables	variables of enclosing scope captured by lambdas in e	§A.1
fv(s)	set of variables	free variables of statement s	§A.1
fv$_e$(e)	set of variables	free variables of expression e	§A.1
Types			
base-type(ℓ), base-type(τ)	type	base type of literal ℓ, or of type τ	§A.2
$\sigma \vee_\Sigma \tau$, $\bigvee_\Sigma \vec{\tau}$	type	larger type if comparable, and union otherwise	§A.2
$\sigma \wedge_\Sigma \tau$	type	greatest type below both	§A.2
elem-type(Σ, τ)	type	type of elements drawn from τ by generator	§A.2
shapes(Σ, τ, H), shapes($\Sigma, \vec{\tau}$)	ordered set of shapes or of shape sequences	shapes of τ once heads in H are excluded; sequences of shapes of $\vec{\tau}$	§A.2
shape-type(k)	type	least type of shape k	§A.2
seq-safe(Σ, Γ, p, τ)	Boolean	tuple patterns in p never checked against lists, list patterns never against tuples	§A.2
Classes and modules			
submods(q)	context	stub for each submodule of q	§A.3
own-fields(ϕ)	sequence of bindings	fields declared by field list ϕ, with their type expressions	§A.3
fields(Σ, C)	sequence of bindings	fields of class C with their types, inherited first	§A.3
ancestors(Σ, C)	set of classes	C and its base classes	§A.3
field-map($\Sigma, C, \vec{u}$)	map from variables	argument supplied for each field of C; undefined unless arguments cover them exactly	§A.3
resolve(ρ, q)	value	value that qualified name q denotes in ρ	§A.3
name(q)	string	string literal naming module q	§A.3
Values			
eq(v, v')	Boolean	structural equality; undefined between unrelated values	§A.4
eq-elems($\vec{v}, \vec{v}\,'$)	Boolean	elementwise equality of value sequences	§A.4
contains(v, v')	Boolean	whether v' is element of v	§A.4
contains-elems($\vec{v}, v$)	Boolean	whether $\vec{v}$ has element equal to v	§A.4
elems(v)	sequence of values	elements of list, tuple or string	§A.4
iter(v)	sequence of values	sequence drawn from v by generator; keys if v is dictionary	§A.4
getitem(v, v')	value or abort	subscripting; IndexError for index out of range, KeyError for missing key, TypeError otherwise	§A.4
result(o)	value or abort	statement outcome as result: returned value, `None` if statement completes without returning, abort unchanged	§A.4
dispatch($\rho, v, \vec{p}, \vec{s}$)	bindings and statement	case taken by `match`	§A.4
update(δ, δ')	dictionary entries	δ updated by δ' in order, later replacing earlier	§A.4
setitem(δ, w, v)	dictionary entries	δ with w bound to v, replacing earlier binding	§A.4

Figure S3: Auxiliary functions

1 Conformance

This document defines PurePy version 0.16.0. PurePy is a subset of Python: every well-formed program is a syntactically valid Python program, and if PurePy gives a run a result, that run produces the same result under Python. Python means Python 3.12 throughout. Chapter 2 gives the abstract syntax; the lexical syntax is that of Python (§2.1). Chapter 3 defines well-formedness, the static part of the definition, and Chapter 4 the operational semantics, which is defined for well-formed programs. A run is *valid* when the semantics assigns it a result; any other run reaches an operation left undefined by the semantics.

Well-formedness is a property of the program, decided by PurePy's static checks. Semantic validity is a property of a run rather than a program, because some of the Python behaviours excluded by PurePy cannot be rejected statically (summarised in the informative Annex B). An implementation conforms if it accepts exactly the well-formed programs and, on every valid run, produces the Python result; on any other run it may abort at the dynamically excluded operation, or simply produce the Python result. The conformance suite (§1.1) organises its tests by verdict: semantically valid PurePy, excluded by PurePy (syntactically, statically or dynamically), or failing under Python.

1.1 Conformance suite

The conformance suite pairs each test with the verdict given by Python itself, and runs on CPython, GraalPy, and Pyodide. [**TODO:** Diagram showing the various kinds of test.] [**TODO:** Reference interpreter, needed to detect dynamically excluded operations (#64).]

2 Syntax

The syntax of PurePy is given in Figures 2.2, 2.3 and 2.4. Figure S1 summarises the metavariables used throughout. A sequence $\vec{x}$ denotes $x_1, \ldots, x_n$ for some n; inside the brackets of a tuple, list, type application or argument list it stands for the comma-separated elements, so $(\vec{e})$ is $(e_1, \ldots, e_n)$. The sequence $x \cdot \vec{x}$ has head x and tail $\vec{x}$, and $\vec{x} +\!\!+ \vec{y}$ is concatenation. Comma-separated elements inside set braces are pairwise distinct, so $\{x_1, \ldots, x_n\}$ has n elements; the set of the elements of a sequence, $\{\vec{x}\}$, assumes no distinctness. A sequence comprehension $(x_i \mid i \in I)$, where the finite index set I carries a total order, denotes the sequence of the x_i in that order. When the order of its elements matters, a set $\{x_1, \ldots, x_n\}$ is *ordered* by position. A comprehension over an ordered set preserves the order of its source; $X \setminus Y$ keeps the surviving elements of X in their order; $X \uplus Y$, defined when X and Y are disjoint, appends the elements of Y to those of X; $X \cup Y$ appends the elements of Y not in X; and a product of ordered sets is ordered lexicographically.

2.1 Lexical syntax

The grammar of PurePy is an abstract syntax, and its metavariables range over values rather than over source text: n is a number and w is a string, not the lexemes that denote them. A number is never negative, since Python writes a negative number as unary minus applied to a literal, and a length or an exit status is an integer. The translation of source text into abstract syntax follows Python, including the indentation that delimits blocks.

2.2 Contexts

Metavariables x, f and c range over *variables*; idiomatically f is used when the variable is expected to denote a function and c when it is expected to denote a class. A *qualified name* q is a non-empty sequence of variables separated by dots (Figure 2.2); a variable is thus a qualified name with a single component.

Definition 2.1 (Context)**.** A context Γ (or Δ) is a finite map from variables to *context entries* θ, which take the following forms:

Entry	Form	Records
variable	τ or ff	type of definitely assigned variable, and ff if not definitely assigned
module stub	$\mathsf{Mod}(q)$	module q, named but not loaded
module reference	$\mathsf{Mod}(q, \Gamma)$	module q with signature Γ
class entry	$\mathsf{Type}(C)$	name denotes class C
predefined name	Predef	name usable only in annotation or as decorator

Definition 2.2 (Class)**.** A class C is a marked name $\widetilde{q.c}$: a module q and a class name c declared in q. A module declares each class name at most once, so the fully qualified name of a class is the identity of the class. A class is also the type of its instances.

Definition 2.3 (Class table)**.** A class table Σ is a finite map from classes to class table entries, with $\Sigma(C)$ of the form $(\vec{x} : \vec{\tau};\ D)$: the distinct fields $\vec{x}$ of C with their declared types $\vec{\tau}$, and its base class D, or $\bot$ if it has none. Every class named in Σ is in $\mathrm{dom}(\Sigma)$, and the relation taking each class of Σ to its base class is acyclic.

The class table is borrowed from Featherweight Java [6]. Module references arise from import statements. A predefined name is a member such as `Callable` or `dataclass` that no expression may mention: it names a type, heads a type form, or decorates a class declaration. The definite-assignment terminology is borrowed from Java [5, §16]: a variable bound to a type is *definitely assigned*, and ff indicates *not definitely assigned*. It is useful to represent contexts as total functions on the set of all variables by extending the codomain with $\bot$ to indicate *undefined*. We write $\Gamma, x : \theta$ for the context that sends x to θ and otherwise agrees with Γ, define $\mathrm{dom}(\Gamma) = \{x \mid \Gamma(x) \neq \bot\}$, and write $\emptyset$ for the context in which all variables are undefined. For a set of variables X, write $\mathsf{ff}(X)$ for the context $\{x : \mathsf{ff} \mid x \in X\}$.

2.2.1 Context operations

Contexts admit a non-commutative monoid of sequential composition $\Gamma \otimes \Delta$ ("override") and an idempotent commutative semigroup of parallel composition $\oplus_\Sigma$ ("merge"), indexed by a class table Σ. Together these form a *right near-semiring* [13]: $\otimes$ right-distributes over $\oplus_\Sigma$, but left distributivity does not hold. Both are defined pointwise (Figure 2.1): override is right-biased and has $\bot$ as both left and right unit; merge is defined only on variable entries, returning ff if either argument is undefined or not definitely assigned, and the join of two types otherwise, so $\bot$ is not a unit.

$\otimes$ (override)

$$\bot \otimes \theta = \theta \otimes \bot = \theta$$
$$\theta \otimes \theta' = \theta' \quad \text{otherwise}$$

$\oplus_\Sigma$ (merge)

$$\bot \oplus_\Sigma \bot = \bot$$
$$\tau \oplus_\Sigma \bot = \bot \oplus_\Sigma \tau = \mathsf{ff}$$
$$\mathsf{ff} \oplus_\Sigma \theta = \theta \oplus_\Sigma \mathsf{ff} = \mathsf{ff}$$
$$\sigma \oplus_\Sigma \tau = \sigma \vee_\Sigma \tau$$

$\boxplus$ (extension)

$$\mathsf{Mod}(q, \Gamma) \boxplus \mathsf{Mod}(q, \Gamma') = \mathsf{Mod}(q, \Gamma \boxplus \Gamma')$$
$$\mathsf{Mod}(q) \boxplus \mathsf{Mod}(q, \Gamma) = \mathsf{Mod}(q, \Gamma) \boxplus \mathsf{Mod}(q) = \mathsf{Mod}(q, \Gamma)$$
$$\bot \boxplus \theta = \theta \boxplus \bot = \theta$$
$$\theta \boxplus \theta' = \theta' \quad \text{otherwise}$$

Figure 2.1: Context operations, defined on context entries and lifted pointwise to contexts: $(\Gamma \cdot \Delta)(x) = \Gamma(x) \cdot \Delta(x)$

A third operation, *context extension* $\Gamma \boxplus \Delta$ (also Figure 2.1), combines module references for the same module recursively, preferring loaded references to stubs, and is otherwise right-biased like override. It is used in processing an import prefix, in which two imports with the same root, such as `a.b` and `a.c`, both bind the root and must combine rather than shadow.

2.3 Expressions and statements

Statements are sequenced by juxtaposition, $s\ s'$; a function body, a branch of a conditional and the body of a case are statements. A *top-level statement* t is a statement or a class declaration, also sequenced by juxtaposition; the body of a module after its imports is a top-level statement, so class declarations occur only at the top level of a module. A *mutual region* $d_1 \ldots d_{n+1}$ is a maximal run of consecutive `def` statements in a sequence: any other statement separates mutual regions. Maximality is not enforced by the grammar in Figure 2.2; the parser is responsible for grouping consecutive `def`s into a single mutual region. The well-formedness rules treat each mutual region as one mutually-recursive binding.

Statements			
s	$::=$	`pass`	(Pass)
	$\mid$	$s\ s'$	(Seq)
	$\mid$	$x = e$	(Assign)
	$\mid$	$x : \psi = e$	(AnnAssign)
	$\mid$	e	(ExprStmt)
	$\mid$	`assert` e	(Assert)
	$\mid$	`assert` e, e'	(AssertMsg)
	$\mid$	`return`	(ReturnNone)
	$\mid$	`return` e	(Return)
	$\mid$	`if` e `:` s `elif` e_1 `:` s_1 $\ldots$ `elif` e_n `:` s_n	(If)
	$\mid$	`if` e `:` s `elif` e_1 `:` s_1 $\ldots$ `elif` e_n `:` s_n `else :` s'	(IfElse)
	$\mid$	`match` e `:` h_1 $\ldots$ h_{n+1}	(Match)
	$\mid$	d_1 $\ldots$ d_{n+1}	(MutualRegion)
Top-level statements			
t	$::=$	s	(Stmt)
	$\mid$	$t\ t'$	(TopSeq)
	$\mid$	`@dataclass class` c `:` ϕ	(Dataclass)
	$\mid$	`@dataclass class` $c(c')$ `:` ϕ	(DataclassExtend)
Imports			
ι	$::=$	`import` q	(Import)
	$\mid$	`from` q `import` $x_1, \ldots, x_{n+1}$	(FromImport)
Cases			
b	$::=$	`case` p `:` s	(Case)
Field lists			
ϕ	$::=$	$x_1 : \psi_1$ $\ldots$ $x_{n+1} : \psi_{n+1}$	(FieldList)
	$\mid$	`pass`	(EmptyFieldList)
Module bodies			
μ	$::=$	ι_1 $\ldots$ ι_n t	(ModuleBody)
Qualified names			
q	$::=$	$x_1 . \cdots . x_{n+1}$	(Name)
Function definitions			
d	$::=$	`def` $f(x_1 : \psi_1, \ldots, x_n : \psi_n)$ `->` ψ `:` s	(Def)

Figure 2.2: Modules and statements

Expressions			
e	$::=$	x	(Var)
	$\mid$	ℓ	(Literal)
	$\mid$	$e(e_1, \ldots, e_n)$	(Call)
	$\mid$	$q(a)$	(Constr)
	$\mid$	$\{e_1 : e'_1, \ldots, e_n : e'_n\}$	(Dict)
	$\mid$	$e.x$	(Attribute)
	$\mid$	$e[e']$	(Subscript)
	$\mid$	$e \odot e'$	(BinOp)
	$\mid$	$\ominus e$	(UnOp)
	$\mid$	e `and` e'	(And)
	$\mid$	e `or` e'	(Or)
	$\mid$	e_1 `if` e `else` e_2	(CondExpr)
	$\mid$	$[e_1, \ldots, e_n]$	(List)
	$\mid$	$(e_1, \ldots, e_n)$	(Tuple)
	$\mid$	$[\, e\ g_1\ \ldots\ g_n]$	(ListComp)
	$\mid$	$\{e : e'\ g_1\ \ldots\ g_n\}$	(DictComp)
	$\mid$	`lambda` $x_1, \ldots, x_n :\ e$	(Lambda)
Comprehension qualifiers			
g	$::=$	`if` e	(Guard)
	$\mid$	`for` x `in` e	(Generator)
Literals			
ℓ	$::=$	n	(Number)
	$\mid$	w	(String)
	$\mid$	`True`	(True)
	$\mid$	`False`	(False)
	$\mid$	`None`	(None)
Argument lists			
a	$::=$	$u_1, \ldots, u_n, x_{n+1}{=}u_{n+1}, \ldots, x_{n+m}{=}u_{n+m}$	(Args)
Patterns			
p	$::=$	ℓ	(PatLiteral)
	$\mid$	$-n$	(PatNegLit)
	$\mid$	x	(PatVar)
	$\mid$	_	(PatWildcard)
	$\mid$	$[p_1, \ldots, p_n]$	(PatList)
	$\mid$	$(p_1, \ldots, p_n)$	(PatTuple)
	$\mid$	$q(a)$	(PatConstr)
	$\mid$	$\{w_1 :\ p_1, \ldots, w_n :\ p_n\}$	(PatDict)
	$\mid$	p `as` x	(PatAs)

Figure 2.3: Expressions and patterns

Primitive types			
ν	$::=$	`object`	(Object)
	$\mid$	`Never`	(Never)
	$\mid$	`None`	(None)
	$\mid$	`bool`	(Bool)
	$\mid$	`int`	(Int)
	$\mid$	`float`	(Float)
	$\mid$	`str`	(Str)
	$\mid$	`Sized`	(Sized)

Type expressions			
ψ	$::=$	ν	(Primitive)
	$\mid$	`list`$[\psi]$	(List)
	$\mid$	`tuple`$[\psi_1, \ldots, \psi_n]$	(Tuple)
	$\mid$	`dict`$[$`str`$, \psi]$	(Dict)
	$\mid$	`Callable`$[[\psi_1, \ldots, \psi_n], \psi]$	(Callable)
	$\mid$	`Literal`$[\ell]$	(Literal)
	$\mid$	`Literal`$[-n]$	(Negative literal)
	$\mid$	q	(ClassName)
	$\mid$	$\widetilde{q}$	(Class)
	$\mid$	$\psi \mid \psi'$	(Union)

Figure 2.4: Type expressions

An *argument list a* (Figure 2.3) is a sequence of positional arguments followed by keyword arguments $x = u$, where an argument u is an expression in a constructor call and a pattern in a constructor pattern. The auxiliary function field-map (§A.3) pairs each field of the class with an argument: the positional arguments fill the leading fields in order, and a keyword argument $x = u$ fills the field x. It is undefined unless every field receives exactly one argument.

Type annotations. Parameters, returns and dataclass fields are annotated with type expressions ψ, and an assignment may be annotated (Figure 2.4). A type expression names a class by a qualified name; a marked name $\widetilde{q}$ is fully qualified, and does not occur in source. A *type* τ is a type expression whose class names are all marked; the type system resolves a type expression into a type (§3.1). We write $\vec{x} : \vec{\psi}$ for the parameter list $x_1 : \psi_1, \ldots, x_n : \psi_n$. The well-formedness rules and operational semantics do not consult annotations, which are given meaning by the type system.

2.3.1 Operators

The metavariable $\odot$ ranges over the supported binary operators other than `and` and `or`, and $\ominus$ over the supported unary operators, both enumerated in Figure 2.5. The operators `and` and `or` are distinct expression forms, since they need not evaluate both operands. The grammar of Figure 2.3 is intentionally ambiguous on operator forms; disambiguation is by the precedence and associativity given in Figure 2.5, which follow Python.

Level	Operators	Arity	Associativity
(loosest) 1	`or`	binary	left
2	`and`	binary	left
3	`not`	unary	—
4	`==` `!=` `<` `<=` `>` `>=` `in` `not in`	binary	left
5	`+` `-`	binary	left
6	`*` `/` `//` `%`	binary	left
7	`+` `-`	unary	—
(tightest) 8	`**`	binary	right

Figure 2.5: Operator precedence

2.4 Modules and programs

A PurePy *program* is a finite map $\mathcal{M}$ from *fully-qualified module names* (Figure 2.2) to *module bodies* μ, of which one is a distinguished *main module* named `__main__`; here name(q) is the string literal naming q, so name(`__main__`) = `"__main__"`. We identify a module with its fully-qualified name, writing "module q" for the module named q. A module body is a sequence of imports $\vec{\imath}$, its *import prefix*, followed by a top-level statement t; since imports are not statements, they can appear neither within a statement nor after it. The statement begins with the assignment `__name__` = name(q), which binds the name of the module when the module loads (§3.5).

A second map $\mathcal{P}$ gives the predefined modules of Figure 2.7, which have no body: $\mathcal{P}$ takes each such name to its environment and context (Definition 2.4). The two domains are disjoint, and their union is prefix-closed: the parent of any module in $\mathrm{dom}(\mathcal{M}) \cup \mathrm{dom}(\mathcal{P})$ is itself in $\mathrm{dom}(\mathcal{M}) \cup \mathrm{dom}(\mathcal{P})$.[1] The maps $\mathcal{M}$ and $\mathcal{P}$ are fixed and ambient in all definitions that follow.

Definition 2.4 (Predefined contexts and environments)**.** $\mathcal{P}$ takes each module q of Figure 2.7 to a pair (Γ, ρ), where Γ binds `__name__` at `str` and each member of q at the context entry listed in Figure 2.6, and ρ binds `__name__` to name(q) and each member to the corresponding value (Figure 2.7). These signatures name types that are themselves members, sometimes of another predefined module, so each is well-formed in the union of the predefined signatures.

Write $q \preceq q'$ (*prefix*) when $q' = q$ or $q' = q.q^\dagger$ for some $q^\dagger$, and $\prec$ for the strict order; the proper prefixes of a module name are its *containing modules*, and it is *contained* in each of them. The metafunctions on modules and classes are defined in §A.3.

[1] A source tree encodes such an $\mathcal{M}$: its modules are the `.py` files and the directories, each named by its path, so the domain is prefix-closed by construction. The statement of a module is the statement sequence of its `.py` file, or of its `__init__.py` for a directory, preceded by the `__name__` assignment; a directory without an `__init__.py` (a namespace package, PEP 420, https://peps.python.org/pep-0420/) contributes that assignment alone.

Module	Member	Entry
builtins	print	Callable[[object], None]
	len	Callable[[Sized], int]
	None	Predef
	object	Predef
	bool	Predef
	int	Predef
	float	Predef
	str	Predef
	list	Predef
	dict	Predef
	tuple	Predef
math	pi	float
	e	float
	sqrt	Callable[[float], float]
	exp	Callable[[float], float]
	log	Callable[[float], float]
	sin	Callable[[float], float]
	cos	Callable[[float], float]
	tan	Callable[[float], float]
	floor	Callable[[float], int]
	ceil	Callable[[float], int]
sys	argv	list[str]
	exit	Callable[[int], Never]
typing	Callable	Predef
	Literal	Predef
	Never	Predef
	Sized	Predef
dataclasses	dataclass	Predef

Figure 2.6: Signature of each predefined module

Module	Member	Value	
`builtins`	`print`		
	`len`	$\mathsf{Prim}(f)$ where	
		$f(v) = \mathsf{val}\ n$	if $\mathrm{iter}(v) = u_1 \cdot \ldots \cdot u_n$
		$f(\vec{u}) = \mathsf{aborts\ TypeError}$	otherwise
	`None`	Opaque(`builtins.None`)	
	`object`	Opaque(`builtins.object`)	
	`bool`	Opaque(`builtins.bool`)	
	`int`	Opaque(`builtins.int`)	
	`float`	Opaque(`builtins.float`)	
	`str`	Opaque(`builtins.str`)	
	`list`	Opaque(`builtins.list`)	
	`dict`	Opaque(`builtins.dict`)	
	`tuple`	Opaque(`builtins.tuple`)	
`math`	`pi`		
	`e`		
	`sqrt`		
	`exp`		
	`log`		
	`sin`		
	`cos`		
	`tan`		
	`floor`		
	`ceil`		
`sys`	`argv`		
	`exit`	$\mathsf{Prim}(f)$ where	
		$f(\varepsilon) = \mathsf{aborts\ SystemExit}(0)$	
		$f(n) = \mathsf{aborts\ SystemExit}(n)$	
`typing`	`Callable`	Opaque(`typing.Callable`)	
	`Literal`	Opaque(`typing.Literal`)	
	`Never`	Opaque(`typing.Never`)	
	`Sized`	Opaque(`typing.Sized`)	
`dataclasses`	`dataclass`	Opaque(`dataclasses.dataclass`)	

Figure 2.7: Predefined modules and the values of their members

3 Well-formedness

PurePy excludes some Python behaviours statically and others only at runtime. Well-formedness is the static part, a judgement on programs defined in this chapter; the operational semantics (Chapter 4) is defined for well-formed programs. A program is well-formed when it is well typed, so the rules of this chapter give each expression a type, each statement a static outcome recording either the assignments made or the fact that the statement definitely returns, and each module a signature. The chapter introduces types and subtyping, then follows the syntax of Chapter 2, taking statements, expressions and patterns, then modules and programs, and Chapter 4 gives each of these a dynamic counterpart.

Three syntactic analyses support the well-formedness rules: assigns(t), the set of variables assigned anywhere in t (without descending into nested function definitions, which have their own scope, so that assigns(t) $=$ dom(Δ) when t : Assigns(Δ)); binds(p), the set of variables introduced by a pattern p, and likewise binds($\vec{g}$) by the generators of a qualifier sequence; and captures(t), the set of variables from the enclosing scope that are captured by lambdas and nested function definitions within t, and likewise captures($\vec{g}$) by lambdas within a qualifier sequence, with companion $\text{captures}_{\mathsf{e}}(e)$ on expressions. All are defined in §A.1.

3.1 Types

$\Gamma \vdash q : \theta$

simple-var
$$\frac{x : \tau \in \Gamma}{\Gamma \vdash x : \tau}$$

simple-module
$$\frac{x : \mathsf{Mod}(q, \Gamma') \in \Gamma}{\Gamma \vdash x : \mathsf{Mod}(q, \Gamma')}$$

simple-class
$$\frac{c : \mathsf{Type}(C) \in \Gamma}{\Gamma \vdash c : \mathsf{Type}(C)}$$

qualified
$$\frac{\Gamma \vdash q : \mathsf{Mod}(q', \Gamma') \qquad \Gamma' \vdash x : \theta}{\Gamma \vdash q.x : \theta}$$

Figure 3.1: Qualified name q resolves to context entry θ under Γ

The rules are intended as a refinement of mypy's: a well-typed PurePy program should be accepted by mypy, and PurePy may reject programs accepted by mypy but not the reverse. Function signatures, assignments and dataclass fields are annotated with type expressions, and a type expression denotes a type under a context (Figure 3.2). Every definition carries an annotation for its parameters and return type, so the type of a definition is given rather than inferred from its body. A class name resolves to a class (Figure 3.1) and every other name must resolve to a predefined name (Figure 2.6). The function base-type(τ) discards the precision of a literal type, taking `Literal[42]` to `int` and leaving every other type alone (§A.2).

3.1.1 Subtyping

Subtyping (Figure 3.3) has every type below itself, a literal type below its base type, and a class below its ancestors, with `object` above every type and `Never` below every type. A union is above each of its members and below every type that is above all of them. A tuple type is covariant, below another

$\Gamma \vdash \psi \rhd \tau$

ty-name
$$\frac{v : \mathsf{Predef} \in \Gamma}{\Gamma \vdash v \rhd v}$$

ty-class
$$\frac{\Gamma \vdash q \ : \ \mathsf{Type}(C)}{\Gamma \vdash q \rhd C}$$

ty-literal
$$\frac{\texttt{Literal} : \mathsf{Predef} \in \Gamma}{\Gamma \vdash \texttt{Literal}[\ell] \rhd \texttt{Literal}[\ell]}$$

ty-list
$$\frac{\texttt{list} : \mathsf{Predef} \in \Gamma \qquad \Gamma \vdash \psi \rhd \tau}{\Gamma \vdash \texttt{list}[\psi] \rhd \texttt{list}[\tau]}$$

ty-dict
$$\frac{\texttt{dict} : \mathsf{Predef} \in \Gamma \qquad \texttt{str} : \mathsf{Predef} \in \Gamma \qquad \Gamma \vdash \psi \rhd \tau}{\Gamma \vdash \texttt{dict}[\texttt{str}, \psi] \rhd \texttt{dict}[\texttt{str}, \tau]}$$

ty-tuple
$$\frac{\texttt{tuple} : \mathsf{Predef} \in \Gamma \qquad \Gamma \vdash \psi_i \rhd \tau_i \quad (\forall i)}{\Gamma \vdash \texttt{tuple}[\vec{\psi}] \rhd \texttt{tuple}[\vec{\tau}]}$$

ty-callable
$$\frac{\texttt{Callable} : \mathsf{Predef} \in \Gamma \qquad \Gamma \vdash \psi_i \rhd \sigma_i \quad (\forall i) \qquad \Gamma \vdash \psi \rhd \tau}{\Gamma \vdash \texttt{Callable}[[\vec{\psi}], \psi] \rhd \texttt{Callable}[[\vec{\sigma}], \tau]}$$

ty-union
$$\frac{\Gamma \vdash \psi \rhd \sigma \qquad \Gamma \vdash \psi' \rhd \tau}{\Gamma \vdash \psi \mid \psi' \rhd \sigma \mid \tau}$$

Figure 3.2: Type expression ψ denotes type τ under Γ

of the same length when each of its components is below the corresponding component. Two types are *equivalent*, $\sigma \equiv \tau$, when each is a subtype of the other; both notations follow Pierce [11]. Lists and dictionaries are related only when their components are equivalent, since both are invariant in mypy and covariance would accept programs rejected by mypy. A callable type is contravariant in its parameters and covariant in its result, following mypy. The types with a length, namely lists, dictionaries, strings and tuples, are below `Sized`. An `int` is a `float`, as PEP 484 requires, so `x: float = 0` is well typed. A `bool` is not an `int`, although Python treats it as one.

Lemma 3.1 (Transitivity of subtyping). *If* $\Sigma \vdash \sigma <: \tau$ *and* $\Sigma \vdash \tau <: \tau'$ *then* $\Sigma \vdash \sigma <: \tau'$.

Subtyping is therefore a preorder: reflexive and transitive but not antisymmetric, because distinct types such as $\sigma \mid \tau$ and $\tau \mid \sigma$ are equivalent.

Two types are *comparable* when either is a subtype of the other. Two comparable types *join* to the larger type and two incomparable types to their union. Two comparable types *meet* to the smaller type; a meet of incomparable types distributes over a union, takes tuples componentwise and callables with parameters joined and results combined using meet, and is `Never` otherwise, since no value belongs to both types. So joining a type with itself gives that type, joining a class with one of its ancestors gives the ancestor, and joining `int` with `str` gives `int | str`. A sequence of types is joined by folding binary join over the sequence, with `Never` as the unit (§A.2).

Lemma 3.2 (Types form a lattice). *Types ordered by subtyping in* Σ*, up to equivalence, form a bounded distributive lattice:* $\sigma \vee_\Sigma \tau$ *is the least upper bound of* σ *and* τ*,* $\sigma \wedge_\Sigma \tau$ *is their greatest lower bound,* `object` *is the greatest type and* `Never` *the least.*

3.2 Expressions

Within an expression the typing judgement gives each subexpression a type, and is bidirectional. *Synthesis* $\Sigma; \Gamma \vdash e \rhd \tau$ reads a type out of an expression (Figures 3.4, 3.5 and 3.6), and *checking* $\Sigma; \Gamma \vdash e \lhd \tau$ tests an expression against a type supplied by its surroundings (Figure 3.7). The division follows the usual recipe for bidirectional systems [2, 3]: introduction forms check, since a lambda's

$\Sigma \vdash \sigma <: \tau$

subty-refl
$$\frac{}{\Sigma \vdash \tau <: \tau}$$

subty-class
$$\frac{D \in \text{ancestors}(\Sigma, C)}{\Sigma \vdash C <: D}$$

subty-never
$$\frac{}{\Sigma \vdash \texttt{Never} <: \tau}$$

subty-object
$$\frac{}{\Sigma \vdash \tau <: \texttt{object}}$$

subty-int-float
$$\frac{}{\Sigma \vdash \texttt{int} <: \texttt{float}}$$

subty-sized
$$\frac{\tau \text{ a list, dictionary, string or tuple type}}{\Sigma \vdash \tau <: \texttt{Sized}}$$

subty-list
$$\frac{\Sigma \vdash \sigma \equiv \tau}{\Sigma \vdash \texttt{list}[\sigma] <: \texttt{list}[\tau]}$$

subty-dict
$$\frac{\Sigma \vdash \sigma \equiv \tau}{\Sigma \vdash \texttt{dict}[\texttt{str}, \sigma] <: \texttt{dict}[\texttt{str}, \tau]}$$

subty-callable
$$\frac{\Sigma \vdash \tau_i <: \sigma_i \quad (\forall i) \qquad \Sigma \vdash \sigma <: \tau}{\Sigma \vdash \texttt{Callable}[[\sigma_1, \ldots, \sigma_n], \sigma] <: \texttt{Callable}[[\tau_1, \ldots, \tau_n], \tau]}$$

subty-tuple
$$\frac{\Sigma \vdash \sigma_i <: \tau_i \quad (\forall i)}{\Sigma \vdash \texttt{tuple}[\sigma_1, \ldots, \sigma_n] <: \texttt{tuple}[\tau_1, \ldots, \tau_n]}$$

subty-literal
$$\frac{\Sigma \vdash \text{base-type}(\ell) <: \tau}{\Sigma \vdash \texttt{Literal}[\ell] <: \tau}$$

subty-union-left
$$\frac{\Sigma \vdash \sigma <: \tau}{\Sigma \vdash \sigma <: \tau \mid \tau'}$$

subty-union-right
$$\frac{\Sigma \vdash \sigma <: \tau'}{\Sigma \vdash \sigma <: \tau \mid \tau'}$$

subty-union
$$\frac{\Sigma \vdash \tau <: \sigma \qquad \Sigma \vdash \tau' <: \sigma}{\Sigma \vdash \tau \mid \tau' <: \sigma}$$

Figure 3.3: Subtyping

parameters cannot be annotated and an empty list does not determine its element type, and elimination forms synthesise, since a call, a subscript or an attribute reference starts from a subterm whose type is known. Type systems that follow the recipe strictly usually rely on an annotation form ($e : \tau$), which synthesises by construction, so a checking form written with an annotation can be used in any context where synthesis is required. Python has no such form, allowing annotations only on parameters, returns and assignments, so PurePy gives each introduction form a synthesis rule wherever its parts determine the type; Swift, Kotlin and Scala take similar approaches for similar reasons [1, 7, 10].

Variables, literals, attribute references, calls, constructor calls, operators and subscripts synthesise. The var rule applies whenever $\Gamma(x)$ is a type, so a variable that is not definitely assigned has no rule. A literal ℓ synthesises the literal type `Literal`$[\ell]$.

The fields of a class C form a sequence $\text{fields}(\Sigma, C)$ of bindings $x : \tau$, one per field: the fields of its base class first, recursively, then the fields declared by C itself, each at its declared type. The order matters, because a constructor call pairs positional arguments with fields by position. A constructor call $q(a)$ pairs each argument with a field using field-map, checks it against that field's type, and synthesises the class C that q resolves to (constr). In an attribute reference $e.x$ the object e must synthesise a class C, and $e.x$ then has the type bound to x in $\text{fields}(\Sigma, C)$ (attr-object). The signature of a predefined module binds `__name__` at `str` and each member at the type listed for it (Figure 2.6, Definition 2.4). A module attribute reference $e.x$ resolves as a qualified name to a definitely assigned member (attr-module). A variable bound to a module or class is not an expression on its own, and occurs only as the head of a qualified name. In a subscript $e[e']$ there is a rule for each container that e can synthesise: a list (subscript-list) or string (subscript-str) with e' of type `int`, and a dictionary with e' of type `str` (subscript-dict). Tuples have three rules: an index that synthesises a literal type gives the component type at that position, counting from the end when the index is negative (subscript-

$\Sigma; \Gamma \vdash e \rhd \tau$

var
$$\frac{x : \tau \in \Gamma}{\Sigma; \Gamma \vdash x \rhd \tau}$$

literal
$$\frac{}{\Sigma; \Gamma \vdash \ell \rhd \texttt{Literal}[\ell]}$$

attr-module
$$\frac{\Gamma \vdash e.x \ : \ \tau}{\Sigma; \Gamma \vdash e.x \rhd \tau}$$

attr-object
$$\frac{\Sigma; \Gamma \vdash e \rhd C \qquad x : \tau \in \text{fields}(\Sigma, C)}{\Sigma; \Gamma \vdash e.x \rhd \tau}$$

call
$$\frac{\Sigma; \Gamma \vdash e \rhd \texttt{Callable}[[\tau_1, \ldots, \tau_n], \tau] \qquad \Sigma; \Gamma \vdash e_i \lhd \tau_i \quad (\forall i)}{\Sigma; \Gamma \vdash e(e_1, \ldots, e_n) \rhd \tau}$$

syn-call-lambda
$$\frac{\Sigma; \Gamma \vdash e_i \rhd \tau_i \quad (\forall i) \qquad \Sigma; \Gamma \otimes \{x_1 : \tau_1, \ldots, x_n : \tau_n\} \vdash e \rhd \tau}{\Sigma; \Gamma \vdash (\texttt{lambda}\ x_1, \ldots, x_n : \ e)(e_1, \ldots, e_n) \rhd \tau}$$

constr
$$\frac{\Gamma \vdash q \ : \ \mathsf{Type}(C) \qquad \Sigma; \Gamma \vdash \text{field-map}(\Sigma, C, a)(x) \lhd \tau \quad (\forall x : \tau \in \text{fields}(\Sigma, C))}{\Sigma; \Gamma \vdash q(a) \rhd C}$$

binop
$$\frac{\Sigma; \Gamma \vdash e \rhd \sigma \qquad \Sigma; \Gamma \vdash e' \rhd \sigma' \qquad \min(\text{overloads}(\Sigma, \odot, \sigma, \sigma')) = ((\tau, \tau'), \tau^\dagger)}{\Sigma; \Gamma \vdash e \odot e' \rhd \tau^\dagger}$$

unop
$$\frac{\Sigma; \Gamma \vdash e \rhd \sigma \qquad \ominus e \neq -n \quad (\forall n) \qquad \min(\text{overloads}(\Sigma, \ominus, \sigma)) = (\tau, \tau^\dagger)}{\Sigma; \Gamma \vdash \ominus e \rhd \tau^\dagger}$$

unop-literal
$$\frac{}{\Sigma; \Gamma \vdash -n \rhd \texttt{Literal}[-n]}$$

syn-cond
$$\frac{\Sigma; \Gamma \vdash e \lhd \texttt{bool} \qquad I = \{i \mid \Sigma; \Gamma \vdash e_i \rhd \sigma_i\} \neq \emptyset \qquad \Sigma; \Gamma \vdash e_j \lhd \bigvee_\Sigma(\sigma_i \mid i \in I) \quad (\forall j \notin I)}{\Sigma; \Gamma \vdash e_1 \ \texttt{if}\ e\ \texttt{else}\ e_2 \rhd \bigvee_\Sigma(\sigma_i \mid i \in I)}$$

and
$$\frac{\Sigma; \Gamma \vdash e \lhd \texttt{bool} \qquad \Sigma; \Gamma \vdash e' \lhd \texttt{bool}}{\Sigma; \Gamma \vdash e\ \texttt{and}\ e' \rhd \texttt{bool}}$$

or
$$\frac{\Sigma; \Gamma \vdash e \lhd \texttt{bool} \qquad \Sigma; \Gamma \vdash e' \lhd \texttt{bool}}{\Sigma; \Gamma \vdash e\ \texttt{or}\ e' \rhd \texttt{bool}}$$

Figure 3.4: Expression synthesis: names, literals, attributes, calls and operators

tuple, subscript-tuple-neg), and any other index of type `int` gives the join of the component types (subscript-tuple-int). A negated numeric literal synthesises the negated literal type (unop-literal), and unop excludes that form, to keep the judgement deterministic. A subscript, attribute reference or call whose subject synthesises a union is typed at each member, with a variable in place of the subject, and synthesises the join of the results (subscript-union, attr-union, call-union); generators need no such rule, because elem-type takes a union to the join of its members' element types. Dunfield and Pfenning [4] state union elimination once, for an arbitrary evaluation context; here union elimination is stated separately for each elimination form.

Lambdas check, as do lists, tuples, dictionaries and comprehensions. An expression that synthesises σ checks against any supertype of σ (check-synth), which applies only when no other rule concludes $\Sigma; \Gamma \vdash e \lhd \tau$, so both judgements are syntax-directed and an expression has at most one derivation in each.

$\Sigma; \Gamma \vdash e \rhd \tau$ (continued)

subscript-list

$$\frac{\Sigma; \Gamma \vdash e \rhd \texttt{list}[\tau] \qquad \Sigma; \Gamma \vdash e' \lhd \texttt{int}}{\Sigma; \Gamma \vdash e[e'] \rhd \tau}$$

subscript-str

$$\frac{\Sigma; \Gamma \vdash e \rhd \texttt{str} \qquad \Sigma; \Gamma \vdash e' \lhd \texttt{int}}{\Sigma; \Gamma \vdash e[e'] \rhd \texttt{str}}$$

subscript-dict

$$\frac{\Sigma; \Gamma \vdash e \rhd \texttt{dict}[\texttt{str}, \tau] \qquad \Sigma; \Gamma \vdash e' \lhd \texttt{str}}{\Sigma; \Gamma \vdash e[e'] \rhd \tau}$$

subscript-tuple

$$\frac{\Sigma; \Gamma \vdash e \rhd \texttt{tuple}[\vec{\tau}] \qquad \Sigma; \Gamma \vdash e' \rhd \texttt{Literal}[n] \qquad 0 \le n < |\vec{\tau}|}{\Sigma; \Gamma \vdash e[e'] \rhd \tau_{n+1}}$$

subscript-tuple-neg

$$\frac{\Sigma; \Gamma \vdash e \rhd \texttt{tuple}[\vec{\tau}] \qquad \Sigma; \Gamma \vdash e' \rhd \texttt{Literal}[-n] \qquad m = |\vec{\tau}| \qquad 0 < n \le m}{\Sigma; \Gamma \vdash e[e'] \rhd \tau_{(m-n)+1}}$$

subscript-tuple-int

$$\frac{\Sigma; \Gamma \vdash e \rhd \texttt{tuple}[\vec{\tau}] \qquad \Sigma; \Gamma \vdash e' \rhd \texttt{int}}{\Sigma; \Gamma \vdash e[e'] \rhd \bigvee_{\Sigma} \vec{\tau}}$$

subscript-union

$$\frac{\Sigma; \Gamma \vdash e \rhd \sigma \mid \sigma' \qquad \Sigma; \Gamma \otimes \{y : \sigma\} \vdash y[e'] \rhd \tau \qquad \Sigma; \Gamma \otimes \{y : \sigma'\} \vdash y[e'] \rhd \tau' \qquad y \notin \mathrm{fv_e}(e')}{\Sigma; \Gamma \vdash e[e'] \rhd \tau \vee_{\Sigma} \tau'}$$

attr-union

$$\frac{\Sigma; \Gamma \vdash e \rhd \sigma \mid \sigma' \qquad \Sigma; \Gamma \otimes \{y : \sigma\} \vdash y.x \rhd \tau \qquad \Sigma; \Gamma \otimes \{y : \sigma'\} \vdash y.x \rhd \tau'}{\Sigma; \Gamma \vdash e.x \rhd \tau \vee_{\Sigma} \tau'}$$

call-union

$$\frac{\Sigma; \Gamma \vdash e \rhd \sigma \mid \sigma' \qquad \Sigma; \Gamma \otimes \{y : \sigma\} \vdash y(\vec{e}) \rhd \tau \qquad \Sigma; \Gamma \otimes \{y : \sigma'\} \vdash y(\vec{e}) \rhd \tau' \qquad y \notin \mathrm{fv_e}(e_i) \quad (\forall i)}{\Sigma; \Gamma \vdash e(\vec{e}) \rhd \tau \vee_{\Sigma} \tau'}$$

Figure 3.5: Expression synthesis: subscripts, and elimination at a union

The forms that synthesise are conditional expressions when a branch synthesises, at the join of the two types when both branches synthesise and otherwise at the synthesising branch's type, the other branch checking against it (syn-cond); tuples when every component synthesises (syn-tuple); comprehensions when the body synthesises (syn-list-comp, syn-dict-comp); and non-empty lists and dictionaries when at least one element synthesises, the remaining elements checking against the join of the base types of the synthesising elements (syn-list, syn-dict). A list, dictionary or comprehension takes the base type of the type synthesised by its elements or body, since `list` and `dict` are invariant and a literal element type would keep the container from matching any other element type. An empty list or dictionary and a lambda have no synthesis rule at all, so checking is the only way to type them. A lambda applied to arguments is the exception, since the arguments give the parameter types, and the call then synthesises the type of the body or checks the body against an expected type (syn-call-lambda, call-lambda). A non-empty list has both, and the checking rule is still needed when the expected

$\Sigma; \Gamma \vdash e \rhd \tau$ (continued)

syn-list

$$\frac{I = \{i \mid \Sigma; \Gamma \vdash e_i \rhd \sigma_i\} \neq \emptyset \qquad \tau = \bigvee_\Sigma(\text{base-type}(\sigma_i) \mid i \in I) \qquad \Sigma; \Gamma \vdash e_j \lhd \tau \quad (\forall j \notin I)}{\Sigma; \Gamma \vdash [e_1, \ldots, e_{n+1}] \rhd \texttt{list}[\tau]}$$

syn-tuple

$$\frac{\Sigma; \Gamma \vdash e_i \rhd \tau_i \quad (\forall i)}{\Sigma; \Gamma \vdash (\vec{e}) \rhd \texttt{tuple}[\vec{\tau}]}$$

syn-dict

$$\frac{\Sigma; \Gamma \vdash e_i \lhd \texttt{str} \quad (\forall i) \qquad I = \{i \mid \Sigma; \Gamma \vdash e'_i \rhd \sigma_i\} \neq \emptyset \qquad \tau = \bigvee_\Sigma(\text{base-type}(\sigma_i) \mid i \in I) \qquad \Sigma; \Gamma \vdash e'_j \lhd \tau \quad (\forall j \notin I)}{\Sigma; \Gamma \vdash \{e_1 : e'_1, \ldots, e_{n+1} : e'_{n+1}\} \rhd \texttt{dict}[\texttt{str}, \tau]}$$

syn-list-comp

$$\frac{\Sigma; \Gamma \vdash \vec{g} : \Delta \qquad \Sigma; \Gamma \otimes \Delta \vdash e \rhd \tau \qquad \text{captures}_e(e) \cap \text{dom}(\Delta) = \emptyset}{\Sigma; \Gamma \vdash [\, e \ \vec{g}] \rhd \texttt{list}[\text{base-type}(\tau)]}$$

syn-dict-comp

$$\frac{\Sigma; \Gamma \vdash \vec{g} : \Delta \qquad \Sigma; \Gamma \otimes \Delta \vdash e \lhd \texttt{str} \qquad \Sigma; \Gamma \otimes \Delta \vdash e' \rhd \tau \qquad \text{captures}_e((e, e')) \cap \text{dom}(\Delta) = \emptyset}{\Sigma; \Gamma \vdash \{e : e' \ \vec{g}\} \rhd \texttt{dict}[\texttt{str}, \text{base-type}(\tau)]}$$

Figure 3.6: Expression synthesis: tuples, lists, dictionaries and comprehensions

element type is wider than the type synthesised by the elements, since `list` is invariant.

Typing a comprehension types its qualifiers, which bind the variables that can be used in the body (Figure 3.7). A generator's variable is bound at elem-type of the type synthesised by the iterated expression (§A.2). A tuple is iterated at the join of the base types of its components. A lambda within a comprehension may not capture a generator variable, a side condition of list-comp, dict-comp and qual-generator. Python closures are late-binding and the comprehension rebinds the variable on each iteration, so the lambda would see the variable's final value; in PurePy a lambda is only able to see the value of the generator variable at each iteration.

3.2.1 Operators

Each operator is overloaded as specified in Figure 3.8. A *resolved overload* $((\tau_1, \tau_2), \tau)$ pairs operand *bounds* with a result type; the judgements of the figure relate operand types to the resolved overloads of an operator.

Definition 3.1 (Overloads at operand types)**.** The set $\text{overloads}(\Sigma, \odot, \sigma_1, \sigma_2)$ consists of the tuples $((\tau_1, \tau_2), \tau)$ such that $\Sigma \vdash \odot(\sigma_1, \sigma_2) : ((\tau_1, \tau_2), \tau)$, ordered by componentwise subtyping in Σ on their bounds (τ_1, τ_2). The unary set $\text{overloads}(\Sigma, \ominus, \sigma)$ is defined in the same way from conclusions $\Sigma \vdash \ominus(\sigma) : (\tau, \tau')$. In both cases min denotes the least element under this order, if one exists.

The binop and unop rules (Figure 3.4) take the result type of $\min(\text{overloads}(\Sigma, \odot, \sigma_1, \sigma_2))$, so `1 + 2` is an `int` and `1 + 2.0` a `float`.

$\Sigma; \Gamma \vdash e \triangleleft \tau$

check-synth

$$\frac{\Sigma; \Gamma \vdash e \triangleright \sigma \qquad \Sigma \vdash \sigma <: \tau \qquad \text{no other rule concludes } \Sigma; \Gamma \vdash e \triangleleft \tau}{\Sigma; \Gamma \vdash e \triangleleft \tau}$$

lambda

$$\frac{\Sigma; \Gamma \otimes \{x_1 : \tau_1, \ldots, x_n : \tau_n\} \vdash e \triangleleft \tau}{\Sigma; \Gamma \vdash \texttt{lambda}\ x_1, \ldots, x_n : e \triangleleft \texttt{Callable}[[\tau_1, \ldots, \tau_n], \tau]}$$

call-lambda

$$\frac{\Sigma; \Gamma \vdash e_i \triangleright \tau_i \quad (\forall i) \qquad \Sigma; \Gamma \otimes \{x_1 : \tau_1, \ldots, x_n : \tau_n\} \vdash e \triangleleft \tau}{\Sigma; \Gamma \vdash (\texttt{lambda}\ x_1, \ldots, x_n : e)(e_1, \ldots, e_n) \triangleleft \tau}$$

list

$$\frac{\Sigma; \Gamma \vdash e_i \triangleleft \tau \quad (\forall i)}{\Sigma; \Gamma \vdash [\vec{e}] \triangleleft \texttt{list}[\tau]}$$

tuple

$$\frac{\Sigma; \Gamma \vdash e_i \triangleleft \tau_i \quad (\forall i)}{\Sigma; \Gamma \vdash (e_1, \ldots, e_n) \triangleleft \texttt{tuple}[\tau_1, \ldots, \tau_n]}$$

dict

$$\frac{\Sigma; \Gamma \vdash e_i \triangleleft \texttt{str} \quad (\forall i) \qquad \Sigma; \Gamma \vdash e'_i \triangleleft \tau \quad (\forall i)}{\Sigma; \Gamma \vdash \{e_1 : e'_1, \ldots, e_n : e'_n\} \triangleleft \texttt{dict}[\texttt{str}, \tau]}$$

cond

$$\frac{\Sigma; \Gamma \vdash e \triangleleft \texttt{bool} \qquad \Sigma; \Gamma \vdash e_1 \triangleleft \tau \qquad \Sigma; \Gamma \vdash e_2 \triangleleft \tau}{\Sigma; \Gamma \vdash e_1\ \texttt{if}\ e\ \texttt{else}\ e_2 \triangleleft \tau}$$

list-comp

$$\frac{\Sigma; \Gamma \vdash \vec{g} : \Delta \qquad \Sigma; \Gamma \otimes \Delta \vdash e \triangleleft \tau \qquad \text{captures}_{\text{e}}(e) \cap \text{dom}(\Delta) = \emptyset}{\Sigma; \Gamma \vdash [\, e\ \vec{g}] \triangleleft \texttt{list}[\tau]}$$

dict-comp

$$\frac{\Sigma; \Gamma \vdash \vec{g} : \Delta \qquad \Sigma; \Gamma \otimes \Delta \vdash e \triangleleft \texttt{str} \qquad \Sigma; \Gamma \otimes \Delta \vdash e' \triangleleft \tau \qquad \text{captures}_{\text{e}}((e, e')) \cap \text{dom}(\Delta) = \emptyset}{\Sigma; \Gamma \vdash \{e : e'\ \vec{g}\} \triangleleft \texttt{dict}[\texttt{str}, \tau]}$$

(a) Expression e checks against type τ

$\Sigma; \Gamma \vdash \vec{g} : \Delta$

quals-nil

$$\frac{}{\Sigma; \Gamma \vdash \varepsilon : \emptyset}$$

qual-guard

$$\frac{\Sigma; \Gamma \vdash e \triangleleft \texttt{bool} \qquad \Sigma; \Gamma \vdash \vec{g} : \Delta}{\Sigma; \Gamma \vdash \texttt{if}\ e \cdot \vec{g} : \Delta}$$

qual-generator

$$\frac{\Sigma; \Gamma \vdash e \triangleright \tau \qquad \text{elem-type}(\Sigma, \tau) = \sigma \qquad \Sigma; \Gamma \otimes \{x : \sigma\} \vdash \vec{g} : \Delta \qquad x \notin \text{captures}(\vec{g})}{\Sigma; \Gamma \vdash \texttt{for}\ x\ \texttt{in}\ e \cdot \vec{g} : \{x : \sigma\} \otimes \Delta}$$

(b) Comprehension qualifiers $\vec{g}$ are well-formed and bind Δ

Figure 3.7: Expression checking

$\Sigma \vdash \odot(\sigma_1, \sigma_2) : ((\tau_1, \tau_2), \tau)$

$\Sigma \vdash \odot(\sigma_1, \sigma_2) : ((\sigma_1, \sigma_2), \texttt{bool})$	if $\odot \in \{\texttt{==}, \texttt{!=}\}$ and σ_1, σ_2 comparable equality types
$\Sigma \vdash \odot(\sigma_1, \texttt{list}[\tau]) : ((\sigma_1, \texttt{list}[\tau]), \texttt{bool})$	if $\odot \in \{\texttt{in}, \texttt{not in}\}$ and σ_1, τ comparable
$\Sigma \vdash \odot(\sigma_1, \texttt{tuple}[\vec{\tau}]) : ((\sigma_1, \texttt{tuple}[\vec{\tau}]), \texttt{bool})$	if $\odot \in \{\texttt{in}, \texttt{not in}\}$ and $\sigma_1, \bigvee_\Sigma \vec{\tau}$ comparable
$\Sigma \vdash \odot(\sigma_1, \sigma_2) : ((\texttt{str}, \texttt{str}), \texttt{bool})$	if $\odot \in \{\texttt{in}, \texttt{not in}\}$ and $\Sigma \vdash \sigma_1, \sigma_2 <: \texttt{str}$
$\Sigma \vdash \odot(\sigma_1, \texttt{dict}[\texttt{str}, \tau]) : ((\texttt{str}, \texttt{dict}[\texttt{str}, \tau]), \texttt{bool})$	if $\odot \in \{\texttt{in}, \texttt{not in}\}$ and $\Sigma \vdash \sigma_1 <: \texttt{str}$
$\Sigma \vdash \odot(\sigma_1, \sigma_2) : ((\texttt{float}, \texttt{float}), \texttt{bool})$	if $\odot \in \{\texttt{<}, \texttt{<=}, \texttt{>}, \texttt{>=}\}$ and $\Sigma \vdash \sigma_1, \sigma_2 <: \texttt{float}$
$\Sigma \vdash \odot(\sigma_1, \sigma_2) : ((\texttt{str}, \texttt{str}), \texttt{bool})$	if $\odot \in \{\texttt{<}, \texttt{<=}, \texttt{>}, \texttt{>=}\}$ and $\Sigma \vdash \sigma_1, \sigma_2 <: \texttt{str}$
$\Sigma \vdash \odot(\sigma_1, \sigma_2) : ((\texttt{int}, \texttt{int}), \texttt{int})$	if $\odot \in \{\texttt{+}, \texttt{-}, \texttt{*}, \texttt{//}, \texttt{\%}\}$ and $\Sigma \vdash \sigma_1, \sigma_2 <: \texttt{int}$
$\Sigma \vdash \odot(\sigma_1, \sigma_2) : ((\texttt{float}, \texttt{float}), \texttt{float})$	if $\odot \in \{\texttt{+}, \texttt{-}, \texttt{*}, \texttt{//}, \texttt{\%}, \texttt{/}, \texttt{**}\}$ and $\Sigma \vdash \sigma_1, \sigma_2 <: \texttt{float}$
$\Sigma \vdash +(\sigma_1, \sigma_2) : ((\texttt{str}, \texttt{str}), \texttt{str})$	if $\Sigma \vdash \sigma_1, \sigma_2 <: \texttt{str}$
$\Sigma \vdash +(\texttt{list}[\sigma], \texttt{list}[\tau]) : ((\texttt{list}[\sigma], \texttt{list}[\tau]), \texttt{list}[\sigma \vee_\Sigma \tau])$	
$\Sigma \vdash +(\texttt{tuple}[\vec{\sigma}], \texttt{tuple}[\vec{\tau}]) : ((\texttt{tuple}[\vec{\sigma}], \texttt{tuple}[\vec{\tau}]), \texttt{tuple}[\vec{\sigma} +\!\!+ \vec{\tau}])$	
$\Sigma \vdash *(\sigma_1, \sigma_2) : ((\texttt{str}, \texttt{int}), \texttt{str})$	if $\Sigma \vdash \sigma_1 <: \texttt{str}$ and $\Sigma \vdash \sigma_2 <: \texttt{int}$
$\Sigma \vdash *(\sigma_1, \sigma_2) : ((\texttt{int}, \texttt{str}), \texttt{str})$	if $\Sigma \vdash \sigma_1 <: \texttt{int}$ and $\Sigma \vdash \sigma_2 <: \texttt{str}$
$\Sigma \vdash *(\texttt{list}[\tau], \sigma_2) : ((\texttt{list}[\tau], \texttt{int}), \texttt{list}[\tau])$	if $\Sigma \vdash \sigma_2 <: \texttt{int}$
$\Sigma \vdash *(\sigma_1, \texttt{list}[\tau]) : ((\texttt{int}, \texttt{list}[\tau]), \texttt{list}[\tau])$	if $\Sigma \vdash \sigma_1 <: \texttt{int}$
$\Sigma \vdash **(\sigma_1, \texttt{Literal}[n]) : ((\texttt{int}, \texttt{Literal}[n]), \texttt{int})$	if $\Sigma \vdash \sigma_1 <: \texttt{int}$ and n an integer
$\Sigma \vdash **(\sigma_1, \texttt{Literal}[-n]) : ((\texttt{int}, \texttt{Literal}[-n]), \texttt{float})$	if $\Sigma \vdash \sigma_1 <: \texttt{int}$ and n a positive integer
$\Sigma \vdash \odot(\sigma_1, \sigma_2) : ((\tau_1, \tau_2), \tau)$	if $\Sigma \vdash \odot(\text{base-type}(\sigma_1), \text{base-type}(\sigma_2)) : ((\tau_1, \tau_2), \tau)$

$\Sigma \vdash \ominus(\sigma) : (\tau, \tau')$

$\texttt{not}(\sigma) : (\texttt{bool}, \texttt{bool})$	if $\Sigma \vdash \sigma <: \texttt{bool}$
$\Sigma \vdash \ominus(\sigma) : (\texttt{int}, \texttt{int})$	if $\ominus \in \{\texttt{+}, \texttt{-}\}$ and $\Sigma \vdash \sigma <: \texttt{int}$
$\Sigma \vdash \ominus(\sigma) : (\texttt{float}, \texttt{float})$	if $\ominus \in \{\texttt{+}, \texttt{-}\}$ and $\Sigma \vdash \sigma <: \texttt{float}$
$\Sigma \vdash \ominus(\sigma) : (\tau, \tau')$	if $\Sigma \vdash \ominus(\text{base-type}(\sigma)) : (\tau, \tau')$

Figure 3.8: Operator overloads

An *equality type* is one for which structural equality is defined:

$\Sigma \vdash \nu$ eq-type	
$\Sigma \vdash \texttt{Literal}[\ell]$ eq-type	
$\Sigma \vdash \texttt{list}[\tau]$ eq-type	if $\Sigma \vdash \tau$ eq-type
$\Sigma \vdash \texttt{dict}[\texttt{str}, \tau]$ eq-type	if $\Sigma \vdash \tau$ eq-type
$\Sigma \vdash \texttt{tuple}[\vec{\tau}]$ eq-type	if $\Sigma \vdash \tau_i$ eq-type $\quad (\forall i)$
$\Sigma \vdash \sigma \mid \tau$ eq-type	if $\Sigma \vdash \sigma$ eq-type and $\Sigma \vdash \tau$ eq-type
$\Sigma \vdash C$ eq-type	if $\Sigma \vdash \tau$ eq-type for each $x : \tau \in \text{fields}(\Sigma, C)$

A callable type is not an equality type, and neither is any type with a callable component, so comparing two functions is rejected statically. The final rule of each table closes the judgement under base-typing, letting `1 == 0` use the equality overload at `int`, since two distinct literal types are not comparable; `2 ** 3` is still an `int`, since the direct instance's bounds ($\texttt{int}, \texttt{Literal}[3]$) lie below the base instance's ($\texttt{float}, \texttt{float}$). For this operator the binop rule is more precise than mypy, whose overloads give `Any` whenever the exponent is not a literal, whereas the least overload resolves to `float`.

Lemma 3.3 (Least overload)**.** *If neither* σ_1 *nor* σ_2 *is a subtype of* `Never` *and* $\mathrm{overloads}(\Sigma, \odot, \sigma_1, \sigma_2)$ *is not empty, then the set has a least element; likewise* $\mathrm{overloads}(\Sigma, \ominus, \sigma)$*.*

3.3 Statements

Every well-typed statement either *definitely returns*, or *definitely assigns* a context Δ, which may be empty. The base cases are that `return e` definitely returns, and `x = e` definitely assigns $\{\texttt{x} : \sigma\}$ for the type σ of `e`. *Static outcomes* R classify the two completions: R is either $\mathsf{Returns}$ (definitely returns) or $\mathsf{Assigns}(\Delta)$ (definitely assigns Δ).

Context operations $\otimes$ and $\oplus_\Sigma$ lift to a right semiring over static outcomes R, with $\mathsf{Assigns}(\varnothing)$ as left and right unit for $\otimes$ and $\mathsf{Returns}$ as unit for $\oplus_\Sigma$ and zero for $\otimes$.

Definition 3.2 (Override and merge on static outcomes)**.**

$$\begin{aligned} \mathsf{Assigns}(\Delta) \oplus_\Sigma \mathsf{Assigns}(\Delta') &= \mathsf{Assigns}(\Delta \oplus_\Sigma \Delta') \\ R \oplus_\Sigma \mathsf{Returns} = \mathsf{Returns} \oplus_\Sigma R &= R \end{aligned}$$

$$\begin{aligned} \mathsf{Assigns}(\Delta) \otimes \mathsf{Assigns}(\Delta') &= \mathsf{Assigns}(\Delta \otimes \Delta') \\ R \otimes \mathsf{Returns} = \mathsf{Returns} \otimes R &= \mathsf{Returns} \end{aligned}$$

The $\mathsf{Returns} \otimes R$ case never arises because statement typing excludes unreachable statements.

The statement judgement $\Sigma; \Gamma; \zeta \vdash s : R$ says that under Γ and optional return type ζ, statement s has static outcome R (Figure 3.9); every rule passes ζ down unchanged. A sequence $s\ s'$ (seq) types s, types s' under Γ overridden by the context assigned by s, and requires that no variable captured by s is assigned in s'. A conditional combines its branches with $\oplus_\Sigma$, merging in $\mathsf{Assigns}(\varnothing)$ when there is no `else` branch. Top-level statements are checked by a separate judgement, $\Sigma; q; \Gamma \vdash t : R \dashv \Sigma'$ (Figure 3.11): a sequence composes in the same way as a statement sequence, a class declaration is a top-level form, and a plain statement is checked with no return type (top-stmt), so a `return` has no derivation at the top level. A top-level statement may not assign a name that an earlier top-level statement bound to a class (top-seq), so a class binding is permanent in its module and no later declaration or assignment can take a declared class's name. A `return` checks its expression against ζ, which must be of the form τ, and a bare `return` requires τ to admit `None`. An annotated assignment checks the expression against the type of the annotation, and an unannotated assignment takes the type synthesised by the expression (assign, assign-unannot). The conditions of `if` and `assert` check against `bool`, and an assertion message against `str`. A class declaration binds the class name to $\mathsf{Type}(C)$, requiring the fields it declares to be distinct and, when it extends another class, the base name to be bound to a class and the additional fields not to repeat an inherited field (class, class-extend). Class declarations also require the predefined name `dataclass` (Figure 2.6) to be in scope; every PurePy class must declare itself a dataclass. Each rule that takes an annotation resolves it to a type (Figure 3.2).

A function definition binds its parameters at their annotated types, plus the names of its mutual region, including the name of the function itself, each at the appropriate `Callable` type; variables assigned in the body start at ff (def), emulating Python's scoping of function bodies, which prefers a locally assigned name to one visible in an enclosing scope. The body is checked with the declared return type, and must have static outcome $\mathsf{Returns}$ unless the function returns `None`, so a function that declares a result definitely returns one.

A `match` scrutinee synthesises a type σ, and its cases $\vec{b}$ are checked using the auxiliary judgement $\Sigma; \Gamma; \zeta; \sigma \vdash L, \vec{b} : S, L'$ (Figure 3.10), which takes the cases in turn: each case's pattern is matched

$\Sigma; \Gamma; \zeta \vdash s : R$

pass

$$\frac{}{\Sigma; \Gamma; \zeta \vdash \texttt{pass} : \mathsf{Assigns}(\emptyset)}$$

assign

$$\frac{\Gamma \vdash \psi \rhd \sigma \qquad \Sigma; \Gamma \vdash e \lhd \sigma \qquad x \notin \mathrm{captures}_{\mathrm{e}}(e)}{\Sigma; \Gamma; \zeta \vdash x : \psi = e : \mathsf{Assigns}(\{x : \sigma\})}$$

assign-unannot

$$\frac{\Sigma; \Gamma \vdash e \rhd \sigma \qquad x \notin \mathrm{captures}_{\mathrm{e}}(e)}{\Sigma; \Gamma; \zeta \vdash x = e : \mathsf{Assigns}(\{x : \sigma\})}$$

expr-stmt

$$\frac{\Sigma; \Gamma \vdash e \rhd \sigma}{\Sigma; \Gamma; \zeta \vdash e : \mathsf{Assigns}(\emptyset)}$$

assert

$$\frac{\Sigma; \Gamma \vdash e \lhd \texttt{bool}}{\Sigma; \Gamma; \zeta \vdash \texttt{assert}\ e : \mathsf{Assigns}(\emptyset)}$$

assert-msg

$$\frac{\Sigma; \Gamma \vdash e \lhd \texttt{bool} \qquad \Sigma; \Gamma \vdash e' \lhd \texttt{str}}{\Sigma; \Gamma; \zeta \vdash \texttt{assert}\ e, e' : \mathsf{Assigns}(\emptyset)}$$

return-none

$$\frac{\Sigma \vdash \texttt{None} <: \tau}{\Sigma; \Gamma; \tau \vdash \texttt{return} : \mathsf{Returns}}$$

return

$$\frac{\Sigma; \Gamma \vdash e \lhd \tau}{\Sigma; \Gamma; \tau \vdash \texttt{return}\ e : \mathsf{Returns}}$$

if

$$\frac{\Sigma; \Gamma \vdash e_i \lhd \texttt{bool} \quad (\forall i) \qquad \Sigma; \Gamma; \zeta \vdash s_i : R_i \quad (\forall i)}{\Sigma; \Gamma; \zeta \vdash \texttt{if}\ e_1 :\ s_1\ \texttt{elif}\ e_2 :\ s_2\ \ldots\ \texttt{elif}\ e_{n+1} :\ s_{n+1} : R_1 \oplus_\Sigma \ldots \oplus_\Sigma R_{n+1}\ \oplus_\Sigma \mathsf{Assigns}(\emptyset)}$$

if-else

$$\frac{\Sigma; \Gamma \vdash e_i \lhd \texttt{bool} \quad (\forall i) \qquad \Sigma; \Gamma; \zeta \vdash s_i : R_i \quad (\forall i)}{\Sigma; \Gamma; \zeta \vdash \texttt{if}\ e_1 :\ s_1\ \texttt{elif}\ e_2 :\ s_2\ \ldots\ \texttt{elif}\ e_{n+1} :\ s_{n+1}\ \texttt{else} :\ s_{n+2} : R_1 \oplus_\Sigma \ldots \oplus_\Sigma R_{n+2}}$$

seq

$$\frac{\Sigma; \Gamma; \zeta \vdash s : \mathsf{Assigns}(\Delta) \qquad \Sigma; \Gamma \otimes \Delta; \zeta \vdash s' : R \qquad \mathrm{captures}(s) \cap \mathrm{assigns}(s') = \emptyset}{\Sigma; \Gamma; \zeta \vdash s\ s' : \mathsf{Assigns}(\Delta) \otimes R}$$

def

$$\frac{\begin{array}{c}\Delta_i = \{\vec{x}_i : \vec{\sigma}_i\} \quad (\forall i) \qquad \Delta' = \{f_1 : \texttt{Callable}[[\vec{\sigma}_1], \tau_1], \ldots, f_{n+1} : \texttt{Callable}[[\vec{\sigma}_{n+1}], \tau_{n+1}]\} \\ \Sigma; \Gamma \otimes \Delta' \otimes \Delta_i \otimes \mathsf{ff}(\mathrm{assigns}(s_i) \setminus \mathrm{dom}(\Delta_i)); \tau_i \vdash s_i : R_i \quad (\forall i) \\ R_i = \mathsf{Returns} \text{ or } \Sigma \vdash \texttt{None} <: \tau_i \quad (\forall i) \qquad \Gamma \vdash \vec{\psi}_i \rhd \vec{\sigma}_i \quad (\forall i) \qquad \Gamma \vdash \psi_i \rhd \tau_i \quad (\forall i)\end{array}}{\Sigma; \Gamma; \zeta \vdash \texttt{def}\ f_1(\vec{x}_1 : \vec{\psi}_1)\ \texttt{->}\ \psi_1 :\ s_1 \cdot \ldots \cdot \texttt{def}\ f_{n+1}(\vec{x}_{n+1} : \vec{\psi}_{n+1})\ \texttt{->}\ \psi_{n+1} :\ s_{n+1} : \mathsf{Assigns}(\Delta')}$$

match

$$\frac{\Sigma; \Gamma \vdash e \rhd \sigma \qquad \Sigma; \Gamma; \zeta; \sigma \vdash \mathrm{shapes}(\Sigma, \sigma, \emptyset), \vec{b} : S, \emptyset}{\Sigma; \Gamma; \zeta \vdash \texttt{match}\ e :\ \vec{b} : S}$$

match-partial

$$\frac{\Sigma; \Gamma \vdash e \rhd \sigma \qquad \Sigma; \Gamma; \zeta; \sigma \vdash \mathrm{shapes}(\Sigma, \sigma, \emptyset), \vec{b} : S, L \qquad L \neq \emptyset}{\Sigma; \Gamma; \zeta \vdash \texttt{match}\ e :\ \vec{b} : S \oplus_\Sigma \mathsf{Assigns}(\emptyset)}$$

Figure 3.9: Given optional return type ζ, statement s has static outcome R

$\Sigma; \Gamma; \zeta; \sigma \vdash L, \vec{b} \;:\; S, L'$

match-case

$$\frac{\text{seq-safe}(\Sigma, \Gamma, p, \sigma) \qquad \Sigma; \Gamma \vdash L, p \rightsquigarrow K, L' \dashv \Delta \qquad \Sigma; \Gamma \otimes \Delta; \zeta \vdash s \;:\; R}{\Sigma; \Gamma; \zeta; \sigma \vdash L, \texttt{case}\ p\texttt{:}\ s \;:\; \mathsf{Assigns}(\Delta) \otimes R, L'}$$

match-cases

$$\frac{\Sigma; \Gamma; \zeta; \sigma \vdash L, b \;:\; S, L' \qquad \Sigma; \Gamma; \zeta; \sigma \vdash L', \vec{b} \;:\; S', L^\dagger}{\Sigma; \Gamma; \zeta; \sigma \vdash L, b \cdot \vec{b} \;:\; S \oplus_\Sigma S', L^\dagger}$$

Figure 3.10: Given optional return type ζ and scrutinee type σ, cases $\vec{b}$ have static outcome S and transform residual L to L'

$\Sigma; q; \Gamma \vdash t \;:\; R \dashv \Sigma'$

top-stmt

$$\frac{\Sigma; \Gamma; \varepsilon \vdash s \;:\; R}{\Sigma; q; \Gamma \vdash s \;:\; R \dashv \Sigma}$$

top-seq

$$\frac{\Sigma; q; \Gamma \vdash t \;:\; \mathsf{Assigns}(\Delta) \dashv \Sigma' \qquad \Sigma'; q; \Gamma \otimes \Delta \vdash t' \;:\; R \dashv \Sigma^\dagger \qquad \text{captures}(t) \cap \text{assigns}(t') = \varnothing \qquad c \notin \text{assigns}(t') \quad (\forall c : \mathsf{Type}(C) \in \Delta)}{\Sigma; q; \Gamma \vdash t\ t' \;:\; \mathsf{Assigns}(\Delta) \otimes R \dashv \Sigma^\dagger}$$

class

$$\frac{\texttt{dataclass} : \mathsf{Predef} \in \Gamma \qquad \text{own-fields}(\phi) = \vec{x} : \vec{\psi},\ \vec{x}\ \text{distinct} \qquad \Gamma \vdash \psi_i \rhd \sigma_i \quad (\forall i)}{\Sigma; q; \Gamma \vdash \texttt{@dataclass class}\ c \;:\; \phi \;:\; \mathsf{Assigns}(\{c : \mathsf{Type}(\widetilde{q.c})\}) \dashv \Sigma, \widetilde{q.c} : (\vec{x} : \vec{\sigma};\ \bot)}$$

class-extend

$$\frac{c' : \mathsf{Type}(D) \in \Gamma \qquad \texttt{dataclass} : \mathsf{Predef} \in \Gamma \qquad \text{own-fields}(\phi) = \vec{x} : \vec{\psi},\ \vec{x}\ \text{distinct} \qquad \Gamma \vdash \psi_i \rhd \sigma_i \quad (\forall i) \qquad \vec{x} \cap \text{dom}(\text{fields}(\Sigma, D)) = \varnothing}{\Sigma; q; \Gamma \vdash \texttt{@dataclass class}\ c(c') \;:\; \phi \;:\; \mathsf{Assigns}(\{c : \mathsf{Type}(\widetilde{q.c})\}) \dashv \Sigma, \widetilde{q.c} : (\vec{x} : \vec{\sigma};\ D)}$$

Figure 3.11: Top-level statement t of module q has static outcome R, extending Σ to Σ'

against the *residual* L, a static over-approximation of the values of σ which remain unmatched by earlier cases, its body is checked under the pattern's bindings, and the outcomes are merged into S. The match rules start the residual from the shapes of σ (Figure 3.12); match applies when the final residual is empty; match-partial applies otherwise, merging in $\mathsf{Assigns}(\varnothing)$ for the values that fall through. Shapes and matching a pattern against the residual are the subject of §3.4.

Each pattern must also be *sequence-safe* at σ (§A.2), because under the evaluation rules a tuple pattern matches only a tuple and a list pattern matches only a list, whereas Python matches both forms of pattern against both kinds of value (Figure 4.12). A match on `list[int] | tuple[int, int]` therefore admits no tuple or list pattern, and a match on `Sized | tuple[int, int]` admits no tuple pattern.

3.4 Pattern matching

To check a `match`, each case `case` p : s is matched in turn against the residual, an ordered set of *shapes* (Figure 3.12). A shape denotes a set of values of a single type, and the residual denotes the values of the scrutinee type which remain unmatched by earlier cases. Matching the case removes from the residual the values matched by p (Figure 3.13); a case that does not match any shape of the residual is unreachable and the match is rejected. The match is exhaustive if the last case leaves the residual empty, and may otherwise fall through (match, match-partial). The usefulness algorithm for ML [9] also checks reachability and exhaustiveness with a single relation. The residual plays the role of the term descriptions of Sestoft [12], and the uncovered sets of Karachalias et al. [8] likewise let the type exclude cases.

3.4.1 Shapes and shape typing

A shape k (Figure 3.12) denotes a set of values of its type shape-type(k) (§A.2). The form $\tau \setminus H$ denotes the values of type τ whose outermost head is not among those in H; the other forms are constructor, tuple, list and dictionary shapes, each with shapes in place of sub-patterns. A list shape also records its element type, which the element shapes do not determine. A constructor shape also carries an excluded set: $C(\vec{k}) \setminus H$ denotes the instances of C, or of any subclass not below a class of H, whose fields, those of C, have shapes $\vec{k}$. A rest shape at a class type denotes the same values as the constructor shape with a rest shape at each field, and is expanded into that form only by splitting (Figure 3.14): a class may refer to itself through its fields, so expanding in advance would not terminate.

A head itself denotes a set of values: a literal its value, a class its instances, and $\texttt{list}_n$ the lists of length n. Head typing $\Sigma \vdash h : \tau$ holds when the values of h lie within τ. Shape typing $\Sigma \vdash k : \tau$ holds when shape-type(k) is a subtype of τ and k is well-formed: each excluded head is typed at the type of the shape (at `str`, for the keys of a dictionary shape) and each component shape at the type of that component.

The auxiliary function shapes(Σ, τ, H) (§A.2) gives the shapes of τ once the heads in H are excluded. A union contributes the shapes of its members and a dictionary type the dictionary form; every other type contributes the single shape $\tau \setminus H$, or no shape at all when H exhausts τ. The rest form therefore never carries a union or dictionary type, and shape-rest states this restriction as a side condition.

3.4.2 Exhaustiveness and reachability

Both reachability and exhaustiveness are determined by the residual left by *matching*. Matching (Figure 3.13) has three forms. The form on a single shape k matches a pattern against that shape, giving matched shapes K, residual L and bindings Δ. A variable pattern binds at shape-type(k), and a pattern named by `as` at the join over the matched shapes, so a case for one alternative of a union leaves a later variable case bound at the remaining members alone. The form on sequences matches the sub-patterns of a tuple, list, dictionary or constructor pattern against the component shapes position by position, and the bindings of the positions, or of the keys of a dictionary pattern, compose by disjoint union, so a pattern that binds a name twice has no derivation. The form on an ordered set of shapes L, used by the match rules, matches each shape of L in turn. Some shapes of L may admit no derivation for the pattern, so pat-shapes takes the shapes of L that do admit a derivation, of which there must be at least one, and passes the remaining shapes into the residual untouched.

A pattern matches a shape directly if the shape has the head tested by the pattern. Otherwise the shape is first *split* by the auxiliary judgement $\Sigma; \Gamma \vdash k, p \rightsquigarrow_{\mathsf{h}} K, L$ (Figure 3.14): the shapes K have the

Heads			
h	$::=$	ℓ	(Literal)
	$\mid$	C	(Class)
	$\mid$	$\texttt{list}_n$	(List of length n)
H	$::=$	$\{h_1, \ldots, h_n\}$	(Excluded heads)

Shapes			
k	$::=$	$\tau \setminus H$	(Rest)
	$\mid$	$C(k_1, \ldots, k_n) \setminus H$	(Constr)
	$\mid$	$(k_1, \ldots, k_n)$	(Tuple)
	$\mid$	$[k_1, \ldots, k_n]_\tau$	(List)
	$\mid$	$\beta \setminus_\tau H$	(Dict)
K, L	$::=$	$\{k_1, \ldots, k_n\}$	(Shape set)

$\Sigma \vdash h : \tau$

head-literal
$$\frac{\Sigma \vdash \texttt{Literal}[\ell] <: \tau}{\Sigma \vdash \ell : \tau}$$

head-class
$$\frac{\Sigma \vdash C <: \tau}{\Sigma \vdash C : \tau}$$

head-list
$$\frac{}{\Sigma \vdash \texttt{list}_n : \texttt{list}[\tau]}$$

$\Sigma \vdash k : \tau$

shape-rest
$$\frac{\Sigma \vdash h : \tau \quad (\forall h \in H) \qquad \tau \text{ neither union nor dictionary type}}{\Sigma \vdash \tau \setminus H : \tau}$$

shape-constr
$$\frac{\text{fields}(\Sigma, C) = \vec{x} : \vec{\sigma} \qquad \Sigma \vdash h : C \quad (\forall h \in H) \qquad \Sigma \vdash k_i : \sigma_i \quad (\forall i)}{\Sigma \vdash C(\vec{k}) \setminus H : C}$$

shape-tuple
$$\frac{\Sigma \vdash k_i : \tau_i \quad (\forall i)}{\Sigma \vdash (\vec{k}) : \texttt{tuple}[\vec{\tau}]}$$

shape-list
$$\frac{\Sigma \vdash k_i : \tau \quad (\forall i)}{\Sigma \vdash [\vec{k}]_\tau : \texttt{list}[\tau]}$$

shape-sub
$$\frac{\Sigma \vdash k : \sigma \qquad \Sigma \vdash \sigma <: \tau}{\Sigma \vdash k : \tau}$$

shape-dict
$$\frac{\Sigma \vdash h : \texttt{str} \quad (\forall h \in H) \qquad \Sigma \vdash \beta(w) : \tau \quad (\forall w \in \text{dom}(\beta)) \qquad \text{dom}(\beta) \cap H = \emptyset}{\Sigma \vdash \beta \setminus_\tau H : \texttt{dict}[\texttt{str}, \tau]}$$

Figure 3.12: Shapes and shape typing

$\Sigma; \Gamma \vdash k, p \rightsquigarrow K, L \dashv \Delta$

pat-var

$$\frac{}{\Sigma; \Gamma \vdash k, x \rightsquigarrow \{k\}, \emptyset \dashv \{x : \text{shape-type}(k)\}}$$

pat-wild

$$\frac{}{\Sigma; \Gamma \vdash k, _ \rightsquigarrow \{k\}, \emptyset \dashv \emptyset}$$

pat-as

$$\frac{\Sigma; \Gamma \vdash k, p \rightsquigarrow K, L \dashv \Delta}{\Sigma; \Gamma \vdash k, p \text{ as } x \rightsquigarrow K, L \dashv \Delta \uplus \{x : \bigvee_\Sigma(\text{shape-type}(k') \mid k' \in K)\}}$$

pat-split

$$\frac{\Sigma; \Gamma \vdash k, p \rightsquigarrow_{\mathsf{h}} K, L \qquad \Sigma; \Gamma \vdash K, p \rightsquigarrow K', L' \dashv \Delta}{\Sigma; \Gamma \vdash k, p \rightsquigarrow K', L \uplus L' \dashv \Delta}$$

pat-literal

$$\frac{}{\Sigma; \Gamma \vdash \texttt{Literal}[\ell] \setminus \emptyset, \ell \rightsquigarrow \{\texttt{Literal}[\ell] \setminus \emptyset\}, \emptyset \dashv \emptyset}$$

pat-tuple

$$\frac{\Sigma; \Gamma \vdash \vec{k}, \vec{p} \rightsquigarrow K, L \dashv \Delta}{\Sigma; \Gamma \vdash (\vec{k}), (\vec{p}) \rightsquigarrow \{(\vec{k}') \mid \vec{k}' \in K\}, \{(\vec{l}) \mid \vec{l} \in L\} \dashv \Delta}$$

pat-list

$$\frac{\Sigma; \Gamma \vdash \vec{k}, \vec{p} \rightsquigarrow K, L \dashv \Delta}{\Sigma; \Gamma \vdash [\vec{k}]_\tau, [\vec{p}] \rightsquigarrow \{[\vec{k}']_\tau \mid \vec{k}' \in K\}, \{[\vec{l}]_\tau \mid \vec{l} \in L\} \dashv \Delta}$$

pat-dict

$$\frac{\vec{w} \text{ distinct} \qquad k_i = \beta(w_i) \quad (\forall i) \qquad \Sigma; \Gamma \vdash \vec{k}, \vec{p} \rightsquigarrow K, L \dashv \Delta}{\Sigma; \Gamma \vdash \beta \setminus_\tau H, \{\vec{w} : \vec{p}\} \rightsquigarrow \{\beta \otimes \{\vec{w} : \vec{k}'\} \setminus_\tau H \mid \vec{k}' \in K\}, \{\beta \otimes \{\vec{w} : \vec{l}\} \setminus_\tau H \mid \vec{l} \in L\} \dashv \Delta}$$

pat-constr

$$\frac{\begin{array}{c}\Gamma \vdash q : \mathsf{Type}(D) \qquad \Sigma \vdash C <: D \\ \text{fields}(\Sigma, D) = \vec{x} : \vec{\sigma} \qquad \text{field-map}(\Sigma, D, a) = \{\vec{x} : \vec{p}\} \qquad p'_i = _ \quad (\forall i) \qquad \Sigma; \Gamma \vdash \vec{k}, \vec{p} +\!\!+ \vec{p}' \rightsquigarrow K, L \dashv \Delta\end{array}}{\Sigma; \Gamma \vdash C(\vec{k}) \setminus H, q(a) \rightsquigarrow \{C(\vec{k}') \setminus H \mid \vec{k}' \in K\}, \{C(\vec{l}) \setminus H \mid \vec{l} \in L\} \dashv \Delta}$$

$\Sigma; \Gamma \vdash \vec{k}, \vec{p} \rightsquigarrow K, L \dashv \Delta$

pat-nil

$$\frac{}{\Sigma; \Gamma \vdash \varepsilon, \varepsilon \rightsquigarrow \{\varepsilon\}, \emptyset \dashv \emptyset}$$

pat-cons

$$\frac{\Sigma; \Gamma \vdash k, p \rightsquigarrow K, L \dashv \Delta \qquad \Sigma; \Gamma \vdash \vec{k}, \vec{p} \rightsquigarrow K', L' \dashv \Delta'}{\Sigma; \Gamma \vdash k \cdot \vec{k}, p \cdot \vec{p} \rightsquigarrow K \times K', (L \times \{\vec{k}\}) \uplus (K \times L') \dashv \Delta \uplus \Delta'}$$

$\Sigma; \Gamma \vdash L, p \rightsquigarrow K, L' \dashv \Delta$

pat-shapes

$$\frac{L' = \{k \in L \mid \Sigma; \Gamma \vdash k, p \rightsquigarrow K_k, L_k \dashv \Delta_k\} \qquad L' \neq \emptyset \qquad \Delta = \{x : \bigvee_\Sigma(\Delta_k(x) \mid k \in L') \mid x \in \text{binds}(p)\}}{\Sigma; \Gamma \vdash L, p \rightsquigarrow \biguplus_{k \in L'} K_k, \biguplus_{k \in L'} L_k \uplus (L \setminus L') \dashv \Delta}$$

Figure 3.13: Pattern p matches shape k, giving matched shapes K, residual L and bindings Δ; lifted to sequences and ordered sets of shapes

$\Sigma;\Gamma \vdash k, p \rightsquigarrow_{\mathsf{h}} K, L$

split-literal

$$\frac{\ell \notin H \qquad \Sigma \vdash \texttt{Literal}[\ell] <: \tau \qquad \tau \text{ not a literal type}}{\Sigma;\Gamma \vdash \tau \setminus H, \ell \rightsquigarrow_{\mathsf{h}} \{\texttt{Literal}[\ell] \setminus \emptyset\}, \text{shapes}(\Sigma, \tau, H \cup \{\ell\})}$$

split-tuple

$$\frac{|\vec{\tau}| = |\vec{p}|}{\Sigma;\Gamma \vdash \texttt{tuple}[\vec{\tau}] \setminus \emptyset, (\vec{p}) \rightsquigarrow_{\mathsf{h}} \{(\vec{k}) \mid \vec{k} \in \text{shapes}(\Sigma, \vec{\tau})\}, \emptyset}$$

split-list

$$\frac{n = |\vec{p}| \qquad \texttt{list}_n \notin H}{\Sigma;\Gamma \vdash \texttt{list}[\tau] \setminus H, [\vec{p}] \rightsquigarrow_{\mathsf{h}} \{[\vec{k}]_\tau \mid \vec{k} \in \text{shapes}(\Sigma, \tau, \emptyset)^n\}, \text{shapes}(\Sigma, \texttt{list}[\tau], H \cup \{\texttt{list}_n\})}$$

split-dict

$$\frac{w_i \notin \text{dom}(\beta) \cup H \qquad w_j \in \text{dom}(\beta) \quad (\forall j < i)}{\Sigma;\Gamma \vdash \beta \setminus_\tau H, \{\vec{w} : \vec{p}\} \rightsquigarrow_{\mathsf{h}} \{\beta \otimes \{w_i : k\} \setminus_\tau H \mid k \in \text{shapes}(\Sigma, \tau, \emptyset)\}, \{\beta \setminus_\tau (H \cup \{w_i\})\}}$$

split-class

$$\frac{\Gamma \vdash q : \mathsf{Type}(C) \qquad \text{ancestors}(\Sigma, C) \cap H = \emptyset \qquad D = \tau \wedge_\Sigma C \qquad \text{fields}(\Sigma, D) = \vec{x} : \vec{\sigma} \qquad H' = \{h \in H \mid \Sigma \vdash h : D\}}{\Sigma;\Gamma \vdash \tau \setminus H, q(a) \rightsquigarrow_{\mathsf{h}} \{D(\vec{k}) \setminus H' \mid \vec{k} \in \text{shapes}(\Sigma, \vec{\sigma})\}, \text{shapes}(\Sigma, \tau, H \cup \{C\})}$$

split-subclass

$$\frac{\Gamma \vdash q : \mathsf{Type}(D) \qquad \Sigma \vdash D <: C \qquad D \neq C \qquad \text{ancestors}(\Sigma, D) \cap H = \emptyset \qquad \text{fields}(\Sigma, D) = \text{fields}(\Sigma, C) +\!\!+ \vec{y} : \vec{\sigma} \qquad H' = \{h \in H \mid \Sigma \vdash h : D\}}{\Sigma;\Gamma \vdash C(\vec{k}) \setminus H, q(a) \rightsquigarrow_{\mathsf{h}} \{D(\vec{k} +\!\!+ \vec{k}') \setminus H' \mid \vec{k}' \in \text{shapes}(\Sigma, \vec{\sigma})\}, \{C(\vec{k}) \setminus (H \cup \{D\})\}}$$

Figure 3.14: Shape k splits at the head tested by p, giving shapes K and residual L

head tested by p, with the shapes of the component types drawn from shapes, and the residual L is the shape k with that head excluded as well. The pat-split rule matches the pattern against K and adds L to the residual. Only the forms with an excluded set split, and no head types at a tuple type, so the excluded set of a tuple shape is empty and split-tuple leaves the empty residual. A constructor pattern whose class lies below a class of the excluded set matches nothing, because an earlier case for the superclass has already matched every instance.

A dictionary shape has a separate form, $\beta \setminus_\tau H$, standing for the dictionaries that hold each key w in the domain of β with a value of shape $\beta(w)$, but that hold no key of H. A dictionary pattern takes its keys in order, with no key occurring twice. A key in neither β nor H splits the shape at that key (split-dict), adding to the residual the dictionaries that lack the key; once every key of the pattern is in β the keys match as a sequence (pat-dict), each key w against $\beta(w)$. The empty dictionary pattern matches every dictionary and leaves the empty residual.

The cases needed to exhaust a match therefore depend on the type of the scrutinee: a union is exhausted by covering each member, a class type by a pattern for the class itself, `bool` by `True` and `False`, and a dictionary type by the empty dictionary pattern, while `int`, `str` and lists have unboundedly many heads and need a variable or wildcard. A literal-typed scrutinee is exhausted by covering its literals, so `x = 42` admits a match with a case for `42` alone, while `x: int = 42` keeps the type open.

Over `tuple[bool, bool]` the cases `(True, _)`, `(_, False)` and `(False, True)` leave the empty residual, although each covers only part of the type.

Lemma 3.4 (Preservation of shape type)**.** *If* $\Sigma \vdash h : \tau$ *for every* $h \in H$ *then* $\Sigma \vdash k : \tau$ *for each* $k \in \text{shapes}(\Sigma, \tau, H)$*. Moreover, if* $\Sigma \vdash k : \tau$ *for each* $k \in L$ *and* $\Sigma; \Gamma \vdash L, p \rightsquigarrow K, L' \dashv \Delta$ *then* $\Sigma \vdash k : \tau$ *for each* $k \in K \uplus L'$*.*

3.5 Modules and programs

Static loading types a module's body and computes its signature Γ, the judgement $\Sigma \vdash_{\mathsf{load}} q : \Gamma \dashv \Sigma'$ (Figure 3.17); it mirrors loading at run time (Chapter 4). The module rule types the body as a top-level statement, and the predefined rule gives a predefined module the signature of Definition 2.4. Both import forms statically load their target, together with its containing modules (Figure 3.16). A plain import also records each of these modules as a member of its parent (the static loads-as judgement), so that the binding created for the root of the target gives attribute access along the whole chain. A from-import also loads any imported name that denotes a submodule. A containing module of the target is exempt from loading first if it is the importing module or one of its containing modules, since at runtime it may still be loading. The exemption applies only to from-imports, because for a plain import the only containing module that could never load first is the importing module itself, and an import naming a module contained in the importing module is excluded outright. Loading a module may thus require other modules to load first; the rules are interpreted inductively, so a program with an import cycle is ill-formed, because a module in a cycle could finish loading only after itself. Program well-formedness (Figure 3.17) requires the main module to be well typed and hence, recursively, every module that the program can load; a module that is never imported is not checked. The rule module types a module body under the context of $\mathcal{P}(\texttt{builtins})$, overridden by the imported context, so the members of `builtins` are in scope unqualified.

$\Sigma \vdash \Gamma, x \ \mathsf{imports}\ \theta \dashv \Sigma'$

$$\text{imp-member}\ \frac{\Gamma \vdash x : \theta}{\Sigma \vdash \Gamma, x \ \mathsf{imports}\ \theta \dashv \Sigma} \qquad \text{imp-submodule}\ \frac{x : \mathsf{Mod}(q) \in \Gamma \qquad \Sigma \vdash_{\mathsf{load}} q : \Delta \dashv \Sigma'}{\Sigma \vdash \Gamma, x \ \mathsf{imports}\ \mathsf{Mod}(q, \Delta) \dashv \Sigma'}$$

Figure 3.15: Name x imported from module signature Γ has context entry θ

The signature Γ produced by static loading (Figure 3.17) comprises the classes and variables defined by a module, together with its intrinsic `__name__` and a stub for each submodule, but not the names that it imports: following PEP 484[1], an imported name is a private dependency, not re-exported as part of the interface. Submodule stubs and the bindings of the module body are disjoint, because the module rules exclude a module that binds a name that is also the name of a submodule.

[1] https://peps.python.org/pep-0484/

$\Sigma; q \vdash \vec{\imath} : \Gamma \dashv \Sigma'$

imports-nil

$$\frac{}{\Sigma; q \vdash \varepsilon : \emptyset \dashv \Sigma}$$

import

$$\frac{q' = x \cdot \vec{x} \qquad q \not\prec q' \qquad \Sigma \vdash_{\mathsf{load}} q' : \Delta \dashv \Sigma_1 \qquad \Sigma_1 \vdash q', \mathsf{Mod}(q', \Delta) \text{ loads-as } \theta \dashv \Sigma_2 \qquad \Sigma_2; q \vdash \vec{\imath} : \Gamma \dashv \Sigma'}{\Sigma; q \vdash \texttt{import } q' \cdot \vec{\imath} : \{x : \theta\} \boxplus \Gamma \dashv \Sigma'}$$

from-import

$$\frac{\Sigma \vdash_{\mathsf{load}} q' : \Delta \dashv \Sigma_0 \qquad q_1^\dagger \prec \cdots \prec q_m^\dagger \text{ the } q^\dagger \prec q' \text{ with } q^\dagger \not\preceq q \qquad \Sigma_{j-1} \vdash_{\mathsf{load}} q_j^\dagger : \Delta_j \dashv \Sigma_j \quad (\forall j) \qquad \Sigma_{m+i-1} \vdash \Delta, x_i \text{ imports } \theta_i \dashv \Sigma_{m+i} \quad (\forall i) \qquad \Sigma_{m+n+1}; q \vdash \vec{\imath} : \Gamma \dashv \Sigma'}{\Sigma; q \vdash \texttt{from } q' \texttt{ import } x_1, \ldots, x_{n+1} \cdot \vec{\imath} : \{x_1 : \theta_1, \ldots, x_{n+1} : \theta_{n+1}\} \boxplus \Gamma \dashv \Sigma'}$$

(a) Import prefix $\vec{\imath}$ of module q produces bindings Γ

$\Sigma \vdash q, \theta \text{ loads-as } \theta' \dashv \Sigma'$

loads-as-simple

$$\frac{}{\Sigma \vdash x, \theta \text{ loads-as } \theta \dashv \Sigma}$$

loads-as-qualified

$$\frac{\Sigma \vdash_{\mathsf{load}} q : \Gamma' \dashv \Sigma_1 \qquad \Sigma_1 \vdash q, \mathsf{Mod}(q, \Gamma' \boxplus \{x : \theta\}) \text{ loads-as } \theta' \dashv \Sigma'}{\Sigma \vdash q.x, \theta \text{ loads-as } \theta' \dashv \Sigma'}$$

(b) Reference θ for module q loads as reference θ' for root of q, statically loading the modules containing q

Figure 3.16: Import typing

$\Sigma \vdash_{\mathsf{load}} q : \Gamma \dashv \Sigma'$

predefined

$$\frac{\mathcal{P}(q) = (\Gamma', \rho)}{\Sigma \vdash_{\mathsf{load}} q : \Gamma' \dashv \Sigma}$$

module

$$\frac{\mathcal{M}(q) = \vec{\imath}\, t \qquad \Sigma; q \vdash \vec{\imath} : \Gamma \dashv \Sigma_1 \qquad \mathcal{P}(\texttt{builtins}) = (\Gamma', \rho) \qquad \Sigma_1; q; \Gamma' \otimes \Gamma \vdash t : \mathsf{Assigns}(\Delta) \dashv \Sigma' \qquad (\mathrm{dom}(\Gamma) \cup \mathrm{dom}(\Delta)) \cap \mathrm{dom}(\mathrm{submods}(q)) = \emptyset}{\Sigma \vdash_{\mathsf{load}} q : \mathrm{submods}(q) \otimes \Delta \dashv \Sigma'}$$

$\vdash \mathcal{M}$

program

$$\frac{\emptyset \vdash_{\mathsf{load}} \texttt{__main__} : \Gamma \dashv \Sigma}{\vdash \mathcal{M}}$$

Figure 3.17: Module typing

4 Operational semantics

4.1 Values and environments

Values, environments, outcomes and results are given in Figure 4.1. A class value $\mathsf{Type}(C)$ is the context entry of the class, and an object $\mathsf{Obj}(C, \rho)$ records its class by name; throughout this section, Σ is the class table produced by static loading of the program (§3.5) and, like $\mathcal{M}$ and $\mathcal{P}$, is ambient. For a statement s, the outcome $\mathsf{assigns}(\rho)$ represents *normal completion* of s, with accumulated environment ρ; the outcome $\mathsf{returns}\ v$ represents *abrupt completion* of s, with return value v; and the outcome $\mathsf{aborts}\ \kappa$ represents abrupt completion of the whole run, with termination kind κ. Composing statements sequentially composes their outcomes according to the following partial monoid, analogous to sequential composition of static outcomes:

$$\begin{aligned} \mathsf{assigns}(\rho) \otimes \mathsf{assigns}(\rho') &= \mathsf{assigns}(\rho \otimes \rho') \\ \mathsf{assigns}(\rho) \otimes \mathsf{returns}\ v &= \mathsf{returns}\ v \\ \mathsf{assigns}(\rho) \otimes \mathsf{aborts}\ \kappa &= \mathsf{aborts}\ \kappa \end{aligned}$$

with $\mathsf{assigns}(\emptyset)$ as left and right unit.

A *termination kind* κ records how a run ends; all but $\mathsf{SystemExit}(n)$ are named after the exceptions raised by the same program under Python in the same circumstances. A primitive $\mathsf{Prim}(f)$ is a predefined function whose meaning f is a partial map from arguments to results (Figure 2.7). Calling $\mathsf{Prim}(f)$ with arguments $\vec{u}$ has result $f(\vec{u})$, and no derivation if $f(\vec{u})$ is undefined.

The rules are given in a monadic metalanguage over results (Figure 4.1), in which $\mathsf{let}\ v \leftarrow r\ \mathsf{in}\ \ldots$ binds the value of a result; for example, eval-binop concludes $\mathsf{let}\ v' \leftarrow r\ \mathsf{in}\ \hat{\odot}(v, v')$, which is $\hat{\odot}(v, v')$ when r is $\mathsf{val}\ v'$, and $\mathsf{aborts}\ \kappa$ when r is $\mathsf{aborts}\ \kappa$. A final premise needs no propagation rule; earlier premises have one each, recording the subexpressions evaluated before the failing one (Figure 4.10). The statement, sequence and module judgements use results of the same form, carrying an outcome, a sequence of values or an environment in place of a value.

4.2 Auxiliary definitions

The operators and the other auxiliary functions the evaluation rules use are defined in §A.4, and module and class resolution in §A.3.

4.3 Expressions and statements

A value has a context entry, and an environment a context, by Figure 4.2. For any context Γ, any $\rho : \Gamma$ and any statement s with $\Gamma \vdash s : R$, the statement evaluation judgement $\rho, s \Rightarrow o$ says that s evaluates to outcome o in ρ (Figures 4.3 and 4.4). Top-level statements are evaluated by a separate judgement, $q, \rho, t \Rightarrow o$ (Figure 4.5), mirroring the static judgement (Figure 3.11): a sequence composes in the same way as a statement sequence, a class declaration binds its name to the class $\widetilde{q.c}$, and a plain statement

Values			
v	$::=$	ℓ	(Literal)
	$\mid$	$[v_1, \ldots, v_n]$	(List)
	$\mid$	$(v_1, \ldots, v_n)$	(Tuple)
	$\mid$	$\{\delta\}$	(Dict)
	$\mid$	$\mathsf{Lam}_\Gamma(\rho, \vec{x}, e)$	(LambdaClosure)
	$\mid$	$\mathsf{Def}_\Gamma(\rho, \vec{d}, i)$	(DefClosure)
	$\mid$	$\mathsf{Obj}(C, \rho)$	(Object)
	$\mid$	$\mathsf{Mod}(q)$	(ModuleStub)
	$\mid$	$\mathsf{Mod}(q, \rho)$	(LoadedModule)
	$\mid$	$\mathsf{Type}(C)$	(Class)
	$\mid$	$\mathsf{Prim}(f)$	(Primitive)
	$\mid$	$\mathsf{Opaque}(q.x)$	(Opaque)
Environments			
ρ	$::=$	$\{x_1 : v_1, \ldots, x_n : v_n\}$	
Dictionary entries			
δ	$::=$	$w_1 : v_1 \cdot \ldots \cdot w_n : v_n$	
Outcomes			
o	$::=$	$\mathsf{assigns}(\rho)$	(Assigns)
	$\mid$	$\mathsf{returns}\ v$	(Returns)
	$\mid$	$\mathsf{aborts}\ \kappa$	(Aborts)
Results			
r	$::=$	$\mathsf{val}\ v$	(Val)
	$\mid$	$\mathsf{aborts}\ \kappa$	(Aborts)
Match results			
m	$::=$	$\mathsf{match}\ \rho$	(Match)
	$\mid$	$\mathsf{no\text{-}match}$	(NoMatch)
Termination kinds			
κ	$::=$	$\mathsf{TypeError} \mid \mathsf{IndexError} \mid \mathsf{KeyError} \mid \mathsf{ZeroDivisionError}$	
	$\mid$	$\mathsf{AssertionError} \mid \mathsf{AssertionError}(w) \mid \mathsf{AttributeError} \mid \mathsf{SystemExit}(n)$	

Figure 4.1: Values, environments, outcomes and results

is evaluated by the statement judgement (eval-top-stmt). Function definitions delimit the scope of sequential-composition effects, so as black boxes functions are always pure. The auxiliary function result (§A.4) reads the outcome of a statement as the result of a call; if the statement completes without returning, the result is val `None`.

For any context Γ, any $\rho : \Gamma$ and any expression e with $\Gamma \vdash e \triangleright \tau$, the expression evaluation judgement $\rho, e \Rightarrow r$ (Figures 4.6 and 4.7) says that e evaluates to result r in ρ, where r is either val v or aborts κ (Figure 4.1). The operators `and` and `or` evaluate their second operand only when the first leaves the result open, and a conditional expression requires its condition to be `True` or `False`, like the `if` statement (eval-if). A comprehension evaluates its body once for each binding produced by its qualifiers, and concatenates the results; a generator draws its bindings from the elements of a value, given by the auxiliary function iter (§A.4): the elements of a list, tuple or string, and the keys of a dictionary, as in Python. Subscripting, given by the auxiliary function getitem (§A.4), indexes a

list, tuple or string by an integer, counting from the end when negative, and a dictionary by a string key, aborting with IndexError, KeyError or TypeError when Python raises the exception of that name; a dictionary subscripted by a value other than a string is undefined. Likewise an attribute reference on a value other than a module or object has no derivation: the attributes of built-in values are undefined.

$\vdash v : \theta$

val-lit
$$\frac{}{\vdash \ell : \texttt{Literal}[\ell]}$$

val-list
$$\frac{\vdash v_i : \tau \quad (\forall i)}{\vdash [\vec{v}] : \texttt{list}[\tau]}$$

val-tuple
$$\frac{\vdash v_i : \tau_i \quad (\forall i)}{\vdash (\vec{v}) : \texttt{tuple}[\vec{\tau}]}$$

val-dict
$$\frac{\vdash v_i : \tau \quad (\forall i) \qquad \vec{w} \text{ distinct}}{\vdash \{\vec{w} : \vec{v}\} : \texttt{dict}[\texttt{str}, \tau]}$$

val-lambda
$$\frac{\vdash \rho : \Gamma \qquad \Gamma \otimes \{\vec{x} : \vec{\sigma}\} \vdash e \lhd \tau}{\vdash \mathsf{Lam}_\Gamma(\rho, \vec{x}, e) : \texttt{Callable}[[\vec{\sigma}], \tau]}$$

val-def
$$\frac{\vdash \rho : \Gamma \qquad d_i = \texttt{def}\ f_i(\vec{x} : \vec{\psi})\ \texttt{->}\ \psi : s \qquad \Gamma \vdash \vec{d} : \mathsf{Assigns}(\Delta)}{\vdash \mathsf{Def}_\Gamma(\rho, \vec{d}, i) : \Delta(f_i)}$$

val-obj
$$\frac{\vdash \rho(x) : \tau \quad (\forall x : \tau \in \mathrm{fields}(\Sigma, C))}{\vdash \mathsf{Obj}(C, \rho) : C}$$

val-prim
$$\frac{\mathcal{P}(q) = (\Gamma, \rho) \qquad \rho(x) = \mathsf{Prim}(f) \qquad \Gamma(x) = \theta}{\vdash \mathsf{Prim}(f) : \theta}$$

val-opaque
$$\frac{}{\vdash \mathsf{Opaque}(q.x) : \mathsf{Predef}}$$

val-mod
$$\frac{q \in \mathrm{dom}(\mathcal{M}) \cup \mathrm{dom}(\mathcal{P})}{\vdash \mathsf{Mod}(q) : \mathsf{Mod}(q)}$$

val-mod-loaded
$$\frac{\vdash \rho : \Gamma}{\vdash \mathsf{Mod}(q, \rho) : \mathsf{Mod}(q, \Gamma)}$$

val-class
$$\frac{C \in \mathrm{dom}(\Sigma)}{\vdash \mathsf{Type}(C) : \mathsf{Type}(C)}$$

$\vdash \rho : \Gamma$

env-nil
$$\frac{}{\vdash \emptyset : \emptyset}$$

env-cons
$$\frac{\vdash v : \theta \qquad \vdash \rho : \Gamma}{\vdash \{x : v\} \otimes \rho : \{x : \theta\} \otimes \Gamma}$$

Figure 4.2: Value and environment typing

4.4 Pattern matching

A `match` statement takes the first case whose pattern matches the value, running its body under the bindings produced by the pattern, and does nothing if no case matches; the auxiliary function dispatch (§A.4) selects the matching case. Whether a pattern matches is determined by the pattern-matching judgement $\rho, p, v \rightsquigarrow m$ (Figure 4.12). A *match result* m is either $\mathsf{match}\ \rho'$, a match, where ρ' binds the variables of the pattern to the values that they matched, or $\mathsf{no\text{-}match}$, a failure to match. When a pattern and a value are not comparable, such as a list pattern against a tuple, there is no derivation and the run has no result. Sub-results compose with $\otimes$, matching sequentially:

$$\mathsf{match}\ \rho \otimes \mathsf{match}\ \rho' = \mathsf{match}\ (\rho \uplus \rho')$$
$$m \otimes \mathsf{no\text{-}match} = \mathsf{no\text{-}match} \otimes m = \mathsf{no\text{-}match}$$

with $\mathsf{match}\ \emptyset$ as left and right unit. A constructor pattern matches any object whose class has the pattern's class among its ancestors, and its sub-patterns bind the fields of the pattern's class. Since an

$\rho, s \Rightarrow o$

eval-pass

$$\frac{}{\rho, \texttt{pass} \Rightarrow \mathsf{assigns}(\emptyset)}$$

eval-assign

$$\frac{\rho, e \Rightarrow r}{\rho, x = e \Rightarrow \mathsf{let}\ v \leftarrow r\ \mathsf{in}\ \mathsf{assigns}(\{x : v\})}$$

eval-assign-annot

$$\frac{\rho, e \Rightarrow r}{\rho, x : \psi = e \Rightarrow \mathsf{let}\ v \leftarrow r\ \mathsf{in}\ \mathsf{assigns}(\{x : v\})}$$

eval-expr-stmt

$$\frac{\rho, e \Rightarrow r}{\rho, e \Rightarrow \mathsf{let}\ v \leftarrow r\ \mathsf{in}\ \mathsf{assigns}(\emptyset)}$$

eval-return

$$\frac{\rho, e \Rightarrow r}{\rho, \texttt{return}\ e \Rightarrow \mathsf{let}\ v \leftarrow r\ \mathsf{in}\ \mathsf{returns}\ v}$$

eval-return-none

$$\frac{}{\rho, \texttt{return} \Rightarrow \mathsf{returns}\ \texttt{None}}$$

eval-def

$$\frac{d_i = \texttt{def}\ f_i(\vec{x}_i : \vec{\psi}_i)\ \texttt{->}\ \psi_i : s_i \quad (\forall i) \qquad v_i = \mathsf{Def}_\Gamma(\rho, \vec{d}, i) \quad (\forall i)}{\rho, \vec{d} \Rightarrow \mathsf{assigns}(\{\vec{f} : \vec{v}\})}$$

eval-assert

$$\frac{\rho, e \Rightarrow \mathsf{val}\ \texttt{True}}{\rho, \texttt{assert}\ e \Rightarrow \mathsf{assigns}(\emptyset)}$$

eval-assert-msg

$$\frac{\rho, e \Rightarrow \mathsf{val}\ \texttt{True}}{\rho, \texttt{assert}\ e, e' \Rightarrow \mathsf{assigns}(\emptyset)}$$

eval-assert-false

$$\frac{\rho, e \Rightarrow \mathsf{val}\ \texttt{False}}{\rho, \texttt{assert}\ e \Rightarrow \mathsf{aborts\ AssertionError}}$$

eval-assert-msg-false

$$\frac{\rho, e \Rightarrow \mathsf{val}\ \texttt{False} \qquad \rho, e' \Rightarrow r}{\rho, \texttt{assert}\ e, e' \Rightarrow \mathsf{let}\ w \leftarrow r\ \mathsf{in}\ \mathsf{aborts\ AssertionError}(w)}$$

eval-if

$$\frac{\rho, e_j \Rightarrow \mathsf{val}\ \texttt{False} \quad (\forall j < i) \qquad \rho, e_i \Rightarrow \mathsf{val}\ \texttt{True} \qquad \rho, s_i \Rightarrow o}{\rho, \texttt{if}\ e_1\ \texttt{:}\ s_1\ \texttt{elif}\ e_2\ \texttt{:}\ s_2\ \ldots\ \texttt{elif}\ e_{n+1}\ \texttt{:}\ s_{n+1} \Rightarrow o}$$

eval-if-none

$$\frac{\rho, e_j \Rightarrow \mathsf{val}\ \texttt{False} \quad (\forall j \in \{1, \ldots, n+1\})}{\rho, \texttt{if}\ e_1\ \texttt{:}\ s_1\ \texttt{elif}\ e_2\ \texttt{:}\ s_2\ \ldots\ \texttt{elif}\ e_{n+1}\ \texttt{:}\ s_{n+1} \Rightarrow \mathsf{assigns}(\emptyset)}$$

eval-if-else

$$\frac{\rho, e_j \Rightarrow \mathsf{val}\ \texttt{False} \quad (\forall j < i) \qquad \rho, e_i \Rightarrow \mathsf{val}\ \texttt{True} \qquad \rho, s_i \Rightarrow o}{\rho, \texttt{if}\ e_1\ \texttt{:}\ s_1\ \texttt{elif}\ e_2\ \texttt{:}\ s_2\ \ldots\ \texttt{elif}\ e_{n+1}\ \texttt{:}\ s_{n+1}\ \texttt{else}\ \texttt{:}\ s_{n+2} \Rightarrow o}$$

eval-else

$$\frac{\rho, e_j \Rightarrow \mathsf{val}\ \texttt{False} \quad (\forall j \in \{1, \ldots, n+1\}) \qquad \rho, s_{n+2} \Rightarrow o}{\rho, \texttt{if}\ e_1\ \texttt{:}\ s_1\ \texttt{elif}\ e_2\ \texttt{:}\ s_2\ \ldots\ \texttt{elif}\ e_{n+1}\ \texttt{:}\ s_{n+1}\ \texttt{else}\ \texttt{:}\ s_{n+2} \Rightarrow o}$$

eval-match

$$\frac{\rho, e \Rightarrow \mathsf{val}\ v \qquad \mathrm{dispatch}(\rho, v, \vec{p}, \vec{s}) = (\rho', s') \qquad \rho \otimes \rho', s' \Rightarrow o}{\rho, \texttt{match}\ e\texttt{:}\ \texttt{case}\ p_1\texttt{:}\ s_1\ \ldots\ \texttt{case}\ p_{n+1}\texttt{:}\ s_{n+1} \Rightarrow \mathsf{assigns}(\rho') \otimes o}$$

eval-seq

$$\frac{\rho, s \Rightarrow \mathsf{assigns}(\rho') \qquad \rho \otimes \rho', s' \Rightarrow o}{\rho, s\ s' \Rightarrow \mathsf{assigns}(\rho') \otimes o}$$

Figure 4.3: Statement s evaluates to outcome o under environment $\rho : \Gamma$

$\rho, s \Rightarrow o$ (continued)

eval-assert-fail

$$\frac{\rho, e \Rightarrow \text{aborts } \kappa}{\rho, \texttt{assert}\ e \Rightarrow \text{aborts } \kappa}$$

eval-assert-msg-fail

$$\frac{\rho, e \Rightarrow \text{aborts } \kappa}{\rho, \texttt{assert}\ e, e' \Rightarrow \text{aborts } \kappa}$$

eval-if-fail

$$\frac{\rho, e_j \Rightarrow \text{val } \texttt{False} \quad (\forall j < i) \qquad \rho, e_i \Rightarrow \text{aborts } \kappa}{\rho, \texttt{if}\ e_1 : s_1\ \texttt{elif}\ e_2 : s_2\ \ldots\ \texttt{elif}\ e_{n+1} : s_{n+1} \Rightarrow \text{aborts } \kappa}$$

eval-if-else-fail

$$\frac{\rho, e_j \Rightarrow \text{val } \texttt{False} \quad (\forall j < i) \qquad \rho, e_i \Rightarrow \text{aborts } \kappa}{\rho, \texttt{if}\ e_1 : s_1\ \texttt{elif}\ e_2 : s_2\ \ldots\ \texttt{elif}\ e_{n+1} : s_{n+1}\ \texttt{else} : s_{n+2} \Rightarrow \text{aborts } \kappa}$$

eval-match-fail

$$\frac{\rho, e \Rightarrow \text{aborts } \kappa}{\rho, \texttt{match}\ e\texttt{:}\ \texttt{case}\ p_1\texttt{:}\ s_1\ \ldots\ \texttt{case}\ p_{n+1}\texttt{:}\ s_{n+1} \Rightarrow \text{aborts } \kappa}$$

eval-seq-fail

$$\frac{\rho, s \Rightarrow \text{aborts } \kappa}{\rho, s\ s' \Rightarrow \text{aborts } \kappa}$$

Figure 4.4: Statement evaluation, continued (aborting cases)

$q, \rho, t \Rightarrow o$

eval-top-stmt

$$\frac{\rho, s \Rightarrow o}{q, \rho, s \Rightarrow o}$$

eval-top-seq

$$\frac{q, \rho, t \Rightarrow \text{assigns}(\rho') \qquad q, \rho \otimes \rho', t' \Rightarrow o}{q, \rho, t\ t' \Rightarrow \text{assigns}(\rho') \otimes o}$$

eval-top-seq-fail

$$\frac{q, \rho, t \Rightarrow \text{aborts } \kappa}{q, \rho, t\ t' \Rightarrow \text{aborts } \kappa}$$

eval-class

$$\frac{}{q, \rho, \texttt{@dataclass class}\ c\ :\ \phi \Rightarrow \text{assigns}(\{c : \text{Type}(\widetilde{q.c})\})}$$

eval-class-extend

$$\frac{}{q, \rho, \texttt{@dataclass class}\ c(c')\ :\ \phi \Rightarrow \text{assigns}(\{c : \text{Type}(\widetilde{q.c})\})}$$

Figure 4.5: Top-level statement t of module q evaluates to outcome o under environment $\rho : \Gamma$

object records its class by name, the class test in eval-pat-constr is a comparison of qualified names.

4.5 Modules and programs

Module and program evaluation are given in Figure 4.15; import evaluation in Figure 4.14; importing names from a loaded module in Figure 4.13. Context extension (Figure 2.1) applies to environments as well, with the same equations on values in place of context entries, except that a loaded module value $\text{Mod}(q, \rho)$ carries an environment whereas the context entry $\text{Mod}(q, \Gamma)$ carries a context. *Loading*, the judgement $q \Rightarrow_{\text{load}} r$, evaluates the import prefix of q and then its statement, mirroring static loading $\vdash_{\text{load}} q : \Gamma$ (Figure 3.17); as with the static rules, the mutual recursion between loading and evaluation is grounded by program well-formedness, which rules out import cycles (§2.4). Loading takes a single module; an import loads the modules containing its target from the root down, each before the module

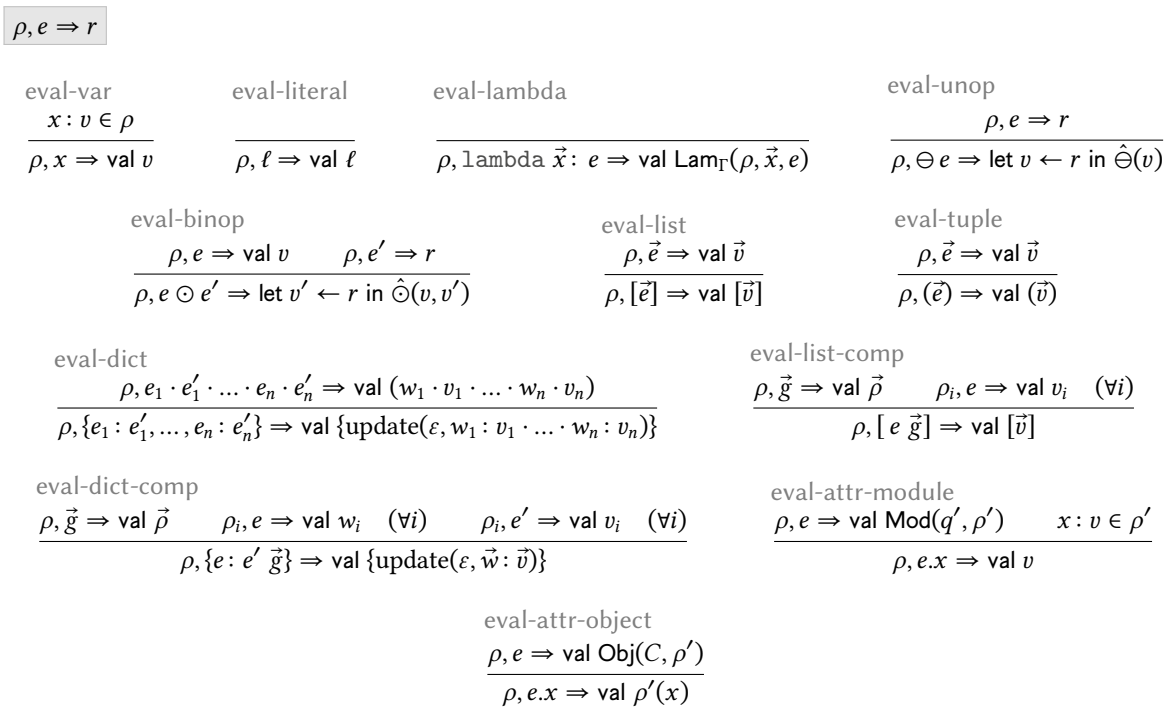


Figure 4.6: Expression e evaluates to result r under environment $\rho : \Gamma$

it contains, mirroring the static rules, by the loads-as judgement for a plain import and as premises of the from-import rule.

Since imports precede statements, every module that a statement can reference is loaded before the statement runs, and loading is deterministic, so the semantics needs no cache of loaded modules. The rule eval-module runs a module body in $\mathcal{P}(\texttt{builtins})$ overridden by the environment of its imports, mirroring module (§3.5). A program evaluates to a termination kind (Figure 4.1): SystemExit(0) if the main module loads, and otherwise the kind it aborted with. SystemExit(n) is a run that exited with status n. An implementation must agree with the semantics on the termination kind of every run; the form in which the kind is reported is not prescribed.

$\rho, e \Rightarrow r$ (continued)

eval-subscript

$$\frac{\rho, e \Rightarrow \mathsf{val}\ v \qquad \rho, e' \Rightarrow r}{\rho, e[e'] \Rightarrow \mathsf{let}\ v' \leftarrow r\ \mathsf{in}\ \mathrm{getitem}(v, v')}$$

eval-call-lambda

$$\frac{\rho, e \Rightarrow \mathsf{val}\ \mathsf{Lam}_\Delta(\rho', \vec{x}, e') \qquad \rho, \vec{e} \Rightarrow \mathsf{val}\ \vec{u} \qquad \rho' \otimes \{\vec{x} : \vec{u}\}, e' \Rightarrow r}{\rho, e(\vec{e}) \Rightarrow r}$$

eval-call-def

$$\frac{\rho, e \Rightarrow \mathsf{val}\ \mathsf{Def}_\Delta(\rho', \vec{d}, j) \qquad d_i = \texttt{def}\ f_i(\vec{x}_i : \vec{\psi}_i)\ \texttt{->}\ \psi_i : s_i \quad (\forall i) \qquad v_i = \mathsf{Def}_\Delta(\rho', \vec{d}, i) \quad (\forall i) \qquad \rho, \vec{e} \Rightarrow \mathsf{val}\ \vec{u} \qquad \rho' \otimes \{\vec{f} : \vec{v}\} \otimes \{\vec{x}_j : \vec{u}\}, s_j \Rightarrow o}{\rho, e(\vec{e}) \Rightarrow \mathrm{result}(o)}$$

eval-call-prim

$$\frac{\rho, e \Rightarrow \mathsf{val}\ \mathsf{Prim}(f) \qquad \rho, \vec{e} \Rightarrow \mathsf{val}\ \vec{u}}{\rho, e(\vec{e}) \Rightarrow f(\vec{u})}$$

eval-and-false

$$\frac{\rho, e \Rightarrow \mathsf{val}\ \texttt{False}}{\rho, e\ \texttt{and}\ e' \Rightarrow \mathsf{val}\ \texttt{False}}$$

eval-and-true

$$\frac{\rho, e \Rightarrow \mathsf{val}\ \texttt{True} \qquad \rho, e' \Rightarrow r}{\rho, e\ \texttt{and}\ e' \Rightarrow r}$$

eval-or-true

$$\frac{\rho, e \Rightarrow \mathsf{val}\ \texttt{True}}{\rho, e\ \texttt{or}\ e' \Rightarrow \mathsf{val}\ \texttt{True}}$$

eval-or-false

$$\frac{\rho, e \Rightarrow \mathsf{val}\ \texttt{False} \qquad \rho, e' \Rightarrow r}{\rho, e\ \texttt{or}\ e' \Rightarrow r}$$

eval-cond-true

$$\frac{\rho, e \Rightarrow \mathsf{val}\ \texttt{True} \qquad \rho, e_1 \Rightarrow r}{\rho, e_1\ \texttt{if}\ e\ \texttt{else}\ e_2 \Rightarrow r}$$

eval-cond-false

$$\frac{\rho, e \Rightarrow \mathsf{val}\ \texttt{False} \qquad \rho, e_2 \Rightarrow r}{\rho, e_1\ \texttt{if}\ e\ \texttt{else}\ e_2 \Rightarrow r}$$

Figure 4.7: Expression evaluation, continued

$\rho, \vec{e} \Rightarrow r$

eval-exprs-nil

$$\frac{}{\rho, \varepsilon \Rightarrow \mathsf{val}\ \varepsilon}$$

eval-exprs-cons

$$\frac{\rho, e \Rightarrow \mathsf{val}\ v \qquad \rho, \vec{e} \Rightarrow r}{\rho, e \cdot \vec{e} \Rightarrow \mathsf{let}\ \vec{v} \leftarrow r\ \mathsf{in}\ \mathsf{val}\ (v \cdot \vec{v})}$$

eval-exprs-fail

$$\frac{\rho, e \Rightarrow \mathsf{aborts}\ \kappa}{\rho, e \cdot \vec{e} \Rightarrow \mathsf{aborts}\ \kappa}$$

Figure 4.8: Expression sequence $\vec{e}$ evaluates left to right to result r under environment $\rho : \Gamma$

$\rho, e \Rightarrow r$ (continued)

eval-binop-fail
$$\frac{\rho, e \Rightarrow \text{aborts } \kappa}{\rho, e \odot e' \Rightarrow \text{aborts } \kappa}$$

eval-and-fail
$$\frac{\rho, e \Rightarrow \text{aborts } \kappa}{\rho, e \texttt{ and } e' \Rightarrow \text{aborts } \kappa}$$

eval-or-fail
$$\frac{\rho, e \Rightarrow \text{aborts } \kappa}{\rho, e \texttt{ or } e' \Rightarrow \text{aborts } \kappa}$$

eval-cond-fail
$$\frac{\rho, e \Rightarrow \text{aborts } \kappa}{\rho, e_1 \texttt{ if } e \texttt{ else } e_2 \Rightarrow \text{aborts } \kappa}$$

eval-attr-fail
$$\frac{\rho, e \Rightarrow \text{aborts } \kappa}{\rho, e.x \Rightarrow \text{aborts } \kappa}$$

eval-subscript-fail
$$\frac{\rho, e \Rightarrow \text{aborts } \kappa}{\rho, e[e'] \Rightarrow \text{aborts } \kappa}$$

eval-call-fail
$$\frac{\rho, e \Rightarrow \text{aborts } \kappa}{\rho, e(\vec{e}) \Rightarrow \text{aborts } \kappa}$$

eval-call-fail-args
$$\frac{\rho, e \Rightarrow \text{val } v \qquad \rho, \vec{e} \Rightarrow \text{aborts } \kappa}{\rho, e(\vec{e}) \Rightarrow \text{aborts } \kappa}$$

eval-list-fail
$$\frac{\rho, \vec{e} \Rightarrow \text{aborts } \kappa}{\rho, [\vec{e}] \Rightarrow \text{aborts } \kappa}$$

eval-tuple-fail
$$\frac{\rho, \vec{e} \Rightarrow \text{aborts } \kappa}{\rho, (\vec{e}) \Rightarrow \text{aborts } \kappa}$$

eval-dict-fail
$$\frac{\rho, e_1 \cdot e'_1 \cdot \ldots \cdot e_n \cdot e'_n \Rightarrow \text{aborts } \kappa}{\rho, \{e_1 : e'_1, \ldots, e_n : e'_n\} \Rightarrow \text{aborts } \kappa}$$

eval-list-comp-fail-quals
$$\frac{\rho, \vec{g} \Rightarrow \text{aborts } \kappa}{\rho, [\, e \ \vec{g}] \Rightarrow \text{aborts } \kappa}$$

eval-list-comp-fail
$$\frac{\rho, \vec{g} \Rightarrow \text{val } \vec{\rho} \qquad \rho_i, e \Rightarrow \text{val } v_i \quad (\forall i < j) \qquad \rho_j, e \Rightarrow \text{aborts } \kappa}{\rho, [\, e \ \vec{g}] \Rightarrow \text{aborts } \kappa}$$

eval-dict-comp-fail-quals
$$\frac{\rho, \vec{g} \Rightarrow \text{aborts } \kappa}{\rho, \{e : e' \ \vec{g}\} \Rightarrow \text{aborts } \kappa}$$

eval-dict-comp-fail-key
$$\frac{\rho, \vec{g} \Rightarrow \text{val } \vec{\rho} \qquad \rho_i, e \Rightarrow \text{val } w_i \quad \rho_i, e' \Rightarrow \text{val } v_i \quad (\forall i < j) \qquad \rho_j, e \Rightarrow \text{aborts } \kappa}{\rho, \{e : e' \ \vec{g}\} \Rightarrow \text{aborts } \kappa}$$

eval-dict-comp-fail-value
$$\frac{\rho, \vec{g} \Rightarrow \text{val } \vec{\rho} \qquad \rho_i, e \Rightarrow \text{val } w_i \quad \rho_i, e' \Rightarrow \text{val } v_i \quad (\forall i < j) \qquad \rho_j, e \Rightarrow \text{val } w_j \qquad \rho_j, e' \Rightarrow \text{aborts } \kappa}{\rho, \{e : e' \ \vec{g}\} \Rightarrow \text{aborts } \kappa}$$

eval-constr-fail
$$\frac{\rho, e_1 \cdot \ldots \cdot e_{n+m} \Rightarrow \text{aborts } \kappa}{\rho, q(e_1, \ldots, e_n, y_1{=}e_{n+1}, \ldots, y_m{=}e_{n+m}) \Rightarrow \text{aborts } \kappa}$$

Figure 4.9: Expression evaluation, continued (propagation from subexpressions)

$\rho, e \Rightarrow r$ (continued)

eval-attr-missing
$$\frac{\rho, e \Rightarrow \text{val Obj}(C, \rho') \qquad x \notin \text{dom}(\rho')}{\rho, e.x \Rightarrow \text{aborts AttributeError}}$$

eval-call-lambda-arity
$$\frac{\rho, e \Rightarrow \text{val Lam}_\Delta(\rho', \vec{x}, e') \qquad \rho, \vec{e} \Rightarrow \text{val } \vec{u} \qquad |\vec{x}| \neq |\vec{u}|}{\rho, e(\vec{e}) \Rightarrow \text{aborts TypeError}}$$

eval-call-def-arity
$$\frac{\rho, e \Rightarrow \text{val Def}_\Delta(\rho', \vec{d}, j) \qquad d_j = \texttt{def } f_j(\vec{x}_j : \vec{\psi}_j) \texttt{ -> } \psi_j : s_j \qquad \rho, \vec{e} \Rightarrow \text{val } \vec{u} \qquad |\vec{x}_j| \neq |\vec{u}|}{\rho, e(\vec{e}) \Rightarrow \text{aborts TypeError}}$$

eval-call-nonfun
$$\frac{\rho, e \Rightarrow \text{val } v \qquad \rho, \vec{e} \Rightarrow \text{val } \vec{u} \qquad v \text{ neither closure nor primitive}}{\rho, e(\vec{e}) \Rightarrow \text{aborts TypeError}}$$

Figure 4.10: Expression evaluation, continued (aborting cases)

$\rho, \vec{g} \Rightarrow r$

eval-quals-nil

$$\frac{}{\rho, \varepsilon \Rightarrow \mathsf{val}\ \rho}$$

eval-qual-guard-true

$$\frac{\rho, e \Rightarrow \mathsf{val}\ \texttt{True} \qquad \rho, \vec{g} \Rightarrow r}{\rho, \texttt{if}\ e \cdot \vec{g} \Rightarrow r}$$

eval-qual-guard-false

$$\frac{\rho, e \Rightarrow \mathsf{val}\ \texttt{False}}{\rho, \texttt{if}\ e \cdot \vec{g} \Rightarrow \mathsf{val}\ \varepsilon}$$

eval-qual-guard-fail

$$\frac{\rho, e \Rightarrow \mathsf{aborts}\ \kappa}{\rho, \texttt{if}\ e \cdot \vec{g} \Rightarrow \mathsf{aborts}\ \kappa}$$

eval-qual-generator

$$\frac{\rho, e \Rightarrow \mathsf{val}\ v \qquad \mathrm{iter}(v) = u_1 \cdot \ldots \cdot u_n \qquad \rho \otimes \{x : u_i\}, \vec{g} \Rightarrow \mathsf{val}\ \vec{\rho}_i \quad (\forall i)}{\rho, \texttt{for}\ x\ \texttt{in}\ e \cdot \vec{g} \Rightarrow \mathsf{val}\ (\vec{\rho}_1 +\!\!+ \ldots +\!\!+ \vec{\rho}_n)}$$

eval-qual-generator-fail

$$\frac{\rho, e \Rightarrow \mathsf{aborts}\ \kappa}{\rho, \texttt{for}\ x\ \texttt{in}\ e \cdot \vec{g} \Rightarrow \mathsf{aborts}\ \kappa}$$

eval-qual-generator-nonseq

$$\frac{\rho, e \Rightarrow \mathsf{val}\ v \qquad \mathrm{iter}(v)\ \mathrm{undefined}}{\rho, \texttt{for}\ x\ \texttt{in}\ e \cdot \vec{g} \Rightarrow \mathsf{aborts}\ \mathsf{TypeError}}$$

eval-qual-generator-fail-body

$$\frac{\rho, e \Rightarrow \mathsf{val}\ v \qquad \mathrm{iter}(v) = u_1 \cdot \ldots \cdot u_n \qquad \rho \otimes \{x : u_i\}, \vec{g} \Rightarrow \mathsf{val}\ \vec{\rho}_i \quad (\forall i < j) \qquad \rho \otimes \{x : u_j\}, \vec{g} \Rightarrow \mathsf{aborts}\ \kappa}{\rho, \texttt{for}\ x\ \texttt{in}\ e \cdot \vec{g} \Rightarrow \mathsf{aborts}\ \kappa}$$

Figure 4.11: Comprehension qualifiers $\vec{g}$ evaluate to result r under environment $\rho : \Gamma$

$\rho, p, v \rightsquigarrow m$

eval-pat-lit
$$\frac{\mathrm{eq}(\ell, v) = \texttt{True}}{\rho, \ell, v \rightsquigarrow \text{match } \emptyset}$$

eval-pat-lit-no
$$\frac{\mathrm{eq}(\ell, v) = \texttt{False}}{\rho, \ell, v \rightsquigarrow \text{no-match}}$$

eval-pat-neg-lit
$$\frac{\mathrm{eq}(-n, v) = \texttt{True}}{\rho, -n, v \rightsquigarrow \text{match } \emptyset}$$

eval-pat-neg-lit-no
$$\frac{\mathrm{eq}(-n, v) = \texttt{False}}{\rho, -n, v \rightsquigarrow \text{no-match}}$$

eval-pat-var
$$\frac{}{\rho, x, v \rightsquigarrow \text{match } \{x : v\}}$$

eval-pat-wild
$$\frac{}{\rho, _, v \rightsquigarrow \text{match } \emptyset}$$

eval-pat-list
$$\frac{\rho, p_i, v_i \rightsquigarrow m_i \quad (\forall i)}{\rho, [p_1, \ldots, p_n], [v_1, \ldots, v_n] \rightsquigarrow m_1 \otimes \ldots \otimes m_n}$$

eval-pat-list-no
$$\frac{v \text{ neither list nor tuple, or list of length} \neq n}{\rho, [p_1, \ldots, p_n], v \rightsquigarrow \text{no-match}}$$

eval-pat-tuple
$$\frac{\rho, p_i, v_i \rightsquigarrow m_i \quad (\forall i)}{\rho, (p_1, \ldots, p_n), (v_1, \ldots, v_n) \rightsquigarrow m_1 \otimes \ldots \otimes m_n}$$

eval-pat-tuple-no
$$\frac{v \text{ neither list nor tuple, or tuple of length} \neq n}{\rho, (p_1, \ldots, p_n), v \rightsquigarrow \text{no-match}}$$

eval-pat-dict
$$\frac{w_i : v_i \in \delta \quad (\forall i) \qquad \rho, p_i, v_i \rightsquigarrow m_i \quad (\forall i)}{\rho, \{w_1 : p_1, \ldots, w_n : p_n\}, \delta \rightsquigarrow m_1 \otimes \ldots \otimes m_n}$$

eval-pat-dict-no
$$\frac{v \text{ not dictionary, or } \delta \text{ with } w_i \notin \mathrm{dom}(\delta) \text{ for some } i}{\rho, \{w_1 : p_1, \ldots, w_n : p_n\}, v \rightsquigarrow \text{no-match}}$$

eval-pat-as
$$\frac{\rho, p, v \rightsquigarrow m}{\rho, p \texttt{ as } x, v \rightsquigarrow m \otimes \text{match } \{x : v\}}$$

eval-pat-constr
$$\frac{\mathrm{resolve}(\rho, q) = \mathsf{Type}(C) \qquad C \in \mathrm{ancestors}(\Sigma, D) \qquad \rho, \text{field-map}(\Sigma, C, \vec{p})(x), \rho'(x) \rightsquigarrow m_x \quad (\forall x \in \mathrm{dom}(\mathrm{fields}(\Sigma, C)))}{\rho, q(\vec{p}), \mathsf{Obj}(D, \rho') \rightsquigarrow \bigotimes_{x \in \mathrm{dom}(\mathrm{fields}(\Sigma, C))} m_x}$$

eval-pat-constr-no
$$\frac{\mathrm{resolve}(\rho, q) = \mathsf{Type}(C) \qquad v \text{ not object, or } \mathsf{Obj}(D, \rho') \text{ with } C \notin \mathrm{ancestors}(\Sigma, D)}{\rho, q(\vec{p}), v \rightsquigarrow \text{no-match}}$$

Figure 4.12: Pattern p matches value v under environment ρ, with result m

ρ, x imports r

eval-imp-member
$$\frac{x : v \in \rho \qquad v \neq \mathsf{Mod}(q') \quad (\forall q')}{\rho, x \text{ imports val } v}$$

eval-imp-submodule
$$\frac{x : \mathsf{Mod}(q') \in \rho \qquad q' \Rightarrow_{\mathsf{load}} \text{val } \rho'}{\rho, x \text{ imports val } \mathsf{Mod}(q', \rho')}$$

eval-imp-submodule-fail
$$\frac{x : \mathsf{Mod}(q') \in \rho \qquad q' \Rightarrow_{\mathsf{load}} \text{aborts } \kappa}{\rho, x \text{ imports aborts } \kappa}$$

Figure 4.13: Name x imported from module environment ρ has result r

$q, \vec{\imath} \Rightarrow r$

eval-imports-nil

$$\frac{}{q, \varepsilon \Rightarrow \mathsf{val}\ \emptyset}$$

eval-import

$$\frac{q' = x \cdot \vec{x} \qquad q' \Rightarrow_{\mathsf{load}} \mathsf{val}\ \rho \qquad q', \mathsf{Mod}(q', \rho)\ \mathsf{loads\text{-}as}\ v \qquad q, \vec{\imath} \Rightarrow r}{q, \texttt{import}\ q' \cdot \vec{\imath} \Rightarrow \mathsf{let}\ \rho' \leftarrow r\ \mathsf{in}\ \mathsf{val}\ (\{x : v\} \boxplus \rho')}$$

eval-import-fail

$$\frac{q' = x \cdot \vec{x} \qquad q' \Rightarrow_{\mathsf{load}} \mathsf{aborts}\ \kappa}{q, \texttt{import}\ q' \cdot \vec{\imath} \Rightarrow \mathsf{aborts}\ \kappa}$$

eval-from-import

$$\frac{q' \Rightarrow_{\mathsf{load}} \mathsf{val}\ \rho \qquad q^\dagger \Rightarrow_{\mathsf{load}} \mathsf{val}\ \rho_{q^\dagger} \quad (\forall q^\dagger \prec q',\ q^\dagger \not\preceq q) \qquad \rho, x_i\ \mathsf{imports}\ \mathsf{val}\ v_i \quad (\forall i) \qquad q, \vec{\imath} \Rightarrow r}{q, \texttt{from}\ q'\ \texttt{import}\ x_1, \ldots, x_{n+1} \cdot \vec{\imath} \Rightarrow \mathsf{let}\ \rho^\dagger \leftarrow r\ \mathsf{in}\ \mathsf{val}\ (\{x_1 : v_1, \ldots, x_{n+1} : v_{n+1}\} \boxplus \rho^\dagger)}$$

eval-from-import-fail

$$\frac{q' \Rightarrow_{\mathsf{load}} \mathsf{aborts}\ \kappa}{q, \texttt{from}\ q'\ \texttt{import}\ x_1, \ldots, x_{n+1} \cdot \vec{\imath} \Rightarrow \mathsf{aborts}\ \kappa}$$

eval-from-import-fail-containing

$$\frac{q' \Rightarrow_{\mathsf{load}} \mathsf{val}\ \rho \qquad q^\dagger \Rightarrow_{\mathsf{load}} \mathsf{aborts}\ \kappa \quad (\mathsf{some}\ q^\dagger \prec q',\ q^\dagger \not\preceq q)}{q, \texttt{from}\ q'\ \texttt{import}\ x_1, \ldots, x_{n+1} \cdot \vec{\imath} \Rightarrow \mathsf{aborts}\ \kappa}$$

eval-from-import-fail-member

$$\frac{\begin{array}{c} q' \Rightarrow_{\mathsf{load}} \mathsf{val}\ \rho \\ q^\dagger \Rightarrow_{\mathsf{load}} \mathsf{val}\ \rho_{q^\dagger} \quad (\forall q^\dagger \prec q',\ q^\dagger \not\preceq q) \qquad \rho, x_i\ \mathsf{imports}\ \mathsf{val}\ v_i \quad (\forall i < j) \qquad \rho, x_j\ \mathsf{imports}\ \mathsf{aborts}\ \kappa \end{array}}{q, \texttt{from}\ q'\ \texttt{import}\ x_1, \ldots, x_{n+1} \cdot \vec{\imath} \Rightarrow \mathsf{aborts}\ \kappa}$$

(a) Import prefix $\vec{\imath}$ of module q evaluates to result r

$q, v\ \mathsf{loads\text{-}as}\ v'$

eval-loads-as-simple

$$\frac{}{x, v\ \mathsf{loads\text{-}as}\ v}$$

eval-loads-as-qualified

$$\frac{q \Rightarrow_{\mathsf{load}} \mathsf{val}\ \rho \qquad q, \mathsf{Mod}(q, \rho \boxplus \{x : v\})\ \mathsf{loads\text{-}as}\ v'}{q.x, v\ \mathsf{loads\text{-}as}\ v'}$$

(b) Reference v for module q loads as reference v' for root of q, loading the modules containing q

Figure 4.14: Import evaluation

$q \Rightarrow_{\mathsf{load}} r$

eval-predefined
$$\frac{\mathcal{P}(q) = (\Gamma, \rho)}{q \Rightarrow_{\mathsf{load}} \mathsf{val}\ \rho}$$

eval-module
$$\frac{\mathcal{M}(q) = \vec{i}\ t \qquad q, \vec{i} \Rightarrow \mathsf{val}\ \rho \qquad \mathcal{P}(\texttt{builtins}) = (\rho', \Gamma') \qquad q, \rho' \otimes \rho, t \Rightarrow \mathsf{assigns}(\rho^\dagger)}{q \Rightarrow_{\mathsf{load}} \mathsf{val}\ (\mathrm{submods}(q) \otimes \rho^\dagger)}$$

eval-module-fail
$$\frac{\mathcal{M}(q) = \vec{i}\ t \qquad q, \vec{i} \Rightarrow \mathsf{aborts}\ \kappa}{q \Rightarrow_{\mathsf{load}} \mathsf{aborts}\ \kappa}$$

eval-module-fail-body
$$\frac{\mathcal{M}(q) = \vec{i}\ t \qquad q, \vec{i} \Rightarrow \mathsf{val}\ \rho \qquad \mathcal{P}(\texttt{builtins}) = (\Gamma', \rho') \qquad q, \rho' \otimes \rho, t \Rightarrow \mathsf{aborts}\ \kappa}{q \Rightarrow_{\mathsf{load}} \mathsf{aborts}\ \kappa}$$

$\mathcal{M} \Rightarrow \kappa$

eval-program
$$\frac{\texttt{__main__} \Rightarrow_{\mathsf{load}} \mathsf{val}\ \rho}{\mathcal{M} \Rightarrow \mathsf{SystemExit}(0)}$$

eval-program-abort
$$\frac{\texttt{__main__} \Rightarrow_{\mathsf{load}} \mathsf{aborts}\ \kappa}{\mathcal{M} \Rightarrow \kappa}$$

Figure 4.15: Module and program evaluation

Annex A

Auxiliary functions (normative)

This annex is normative. It defines the auxiliary functions used by the well-formedness rules (Chapter 3) and the operational semantics (Chapter 4). When more than one case applies, the first defines the value, so a function with overlapping cases is still determined: two equivalent types join to the second of them, for instance.

A.1 Syntactic analyses

The syntactic analyses introduced in Chapter 3 are defined by their principal cases; the remaining cases are stated with each.

assigns(t) — *set of variables*

Variables assigned anywhere in t, not descending into nested functions, which have their own scope. It is empty on the statements not shown and is the union over the branches of a conditional, and on a module body $\vec{\imath}\ t$ acts on t.

$$\begin{aligned}
\text{assigns}(x = e) = \text{assigns}(x : \psi = e) &= \{x\} \\
\text{assigns}(t\ t') &= \text{assigns}(t) \cup \text{assigns}(t') \\
\text{assigns}(\vec{d}) &= \{\vec{f}\} \quad \text{where } d_i = \texttt{def}\ f_i(\vec{x}_i : \vec{\psi}_i)\ \texttt{->}\ \psi_i : s_i \\
\text{assigns}(\texttt{match}\ e : \texttt{case}\ p_1 : b_1\ \ldots\ \texttt{case}\ p_{n+1} : b_{n+1}) &= \bigcup_i \left(\text{binds}(p_i) \cup \text{assigns}(b_i)\right) \\
\text{assigns}(\texttt{@dataclass class}\ c\ :\ \phi) &= \{c\} \\
\text{assigns}(\texttt{@dataclass class}\ c(c')\ :\ \phi) &= \{c\}
\end{aligned}$$

binds(p) — *set of variables*

Variables introduced by pattern p, or by the generators of qualifier sequence $\vec{g}$. On the cases not shown it recurses structurally, taking the union over the immediate subpatterns.

$$\begin{aligned}
\text{binds}(x) &= \{x\} \\
\text{binds}(p\ \texttt{as}\ x) &= \text{binds}(p) \cup \{x\} \\
\text{binds}(\varepsilon) &= \emptyset \\
\text{binds}(\texttt{if}\ e \cdot \vec{g}) &= \text{binds}(\vec{g}) \\
\text{binds}(\texttt{for}\ x\ \texttt{in}\ e \cdot \vec{g}) &= \{x\} \cup \text{binds}(\vec{g})
\end{aligned}$$

captures(t) — *set of variables*

Variables of the enclosing scope captured by lambdas and nested function definitions within t, or by lambdas within the qualifier sequence $\vec{g}$ of a comprehension. On the cases not shown it takes the union of captures_e over the subexpressions of t and of captures over its sub-statements, and on a module body agrees with assigns.

$$\begin{aligned}
\text{captures}(t\ t') &= \text{captures}(t) \cup \text{captures}(t') \\
\text{captures}(\vec{d}) &= \bigcup_i \left(\text{fv}(b_i) \setminus (\{\vec{x}_i\} \cup \text{assigns}(b_i))\right) \setminus \{\vec{f}\} \\
&\quad \text{where } d_i = \texttt{def}\ f_i(\vec{x}_i : \vec{\psi}_i)\ \texttt{->}\ \psi_i : s_i \\
\text{captures}(\texttt{match}\ e : \texttt{case}\ p_1 : b_1\ \ldots\ \texttt{case}\ p_{n+1} : b_{n+1}) &= \text{captures}_\text{e}(e) \cup \bigcup_i \text{captures}(b_i) \\
\text{captures}(\varepsilon) &= \emptyset \\
\text{captures}(\texttt{if}\ e \cdot \vec{g}) &= \text{captures}_\text{e}(e) \cup \text{captures}(\vec{g}) \\
\text{captures}(\texttt{for}\ x\ \texttt{in}\ e \cdot \vec{g}) &= \text{captures}_\text{e}(e) \cup (\text{captures}(\vec{g}) \setminus \{x\})
\end{aligned}$$

$\text{captures}_\text{e}(e)$ — *set of variables*

Variables of the enclosing scope captured by lambdas within expression e; the generator variables of a comprehension are local to it. The subscript distinguishes this from captures(t), since an expression statement is both an expression and a statement. On the cases not shown it recurses structurally, taking the union over the immediate subexpressions.

$$\begin{aligned}
\text{captures}_\text{e}(\texttt{lambda}\ \vec{x} :\ e) &= \text{fv}_\text{e}(e) \setminus \{\vec{x}\} \\
\text{captures}_\text{e}([\ e\ \vec{g}\ ]) &= \text{captures}(\vec{g}) \cup (\text{captures}_\text{e}(e) \setminus \text{binds}(\vec{g})) \\
\text{captures}_\text{e}(\{e : e'\ \vec{g}\}) &= \text{captures}(\vec{g}) \cup (\text{captures}_\text{e}((e, e')) \setminus \text{binds}(\vec{g}))
\end{aligned}$$

fv(s) — *set of variables*

Free variables of statement s, defined in the usual way.

$\text{fv}_\text{e}(e)$ — *set of variables*

Free variables of expression e, defined in the usual way.

A.2 Types

The functions on types and shapes introduced in §3.1, each defined by its cases.

base-type(ℓ), base-type(τ)	*type*

Base type of literal ℓ, and of type τ that type with a literal type replaced by the base type of its literal.

$$
\begin{aligned}
\text{base-type}(n) &= \texttt{int} && \text{if } n \text{ is an integer literal} \\
\text{base-type}(n) &= \texttt{float} && \text{if } n \text{ is a float literal} \\
\text{base-type}(w) &= \texttt{str} \\
\text{base-type}(\texttt{True}) = \text{base-type}(\texttt{False}) &= \texttt{bool} \\
\text{base-type}(\texttt{None}) &= \texttt{None} \\
\text{base-type}(\texttt{Literal}[\ell]) &= \text{base-type}(\ell) \\
\text{base-type}(\tau) &= \tau && \text{otherwise}
\end{aligned}
$$

$\sigma \vee_\Sigma \tau, \bigvee_\Sigma \vec{\tau}$	*type*

Larger of two comparable types, and their union otherwise. A sequence joins its first type with the join of the rest, with `Never` as the unit.

$$
\begin{aligned}
\sigma \vee_\Sigma \tau &= \tau && \text{if } \Sigma \vdash \sigma <: \tau \\
\sigma \vee_\Sigma \tau &= \sigma && \text{if } \Sigma \vdash \tau <: \sigma \text{ and } \Sigma \nvdash \sigma \equiv \tau \\
\sigma \vee_\Sigma \tau &= \sigma \mid \tau && \text{otherwise} \\
\bigvee_\Sigma \varepsilon &= \texttt{Never} \\
\bigvee_\Sigma \sigma \cdot \vec{\tau} &= \sigma \vee_\Sigma \bigvee_\Sigma \vec{\tau}
\end{aligned}
$$

$\sigma \wedge_\Sigma \tau$	*type*

Cases after the first two apply only when σ and τ are incomparable.

$$
\begin{aligned}
\sigma \wedge_\Sigma \tau &= \sigma && \text{if } \Sigma \vdash \sigma <: \tau \\
\sigma \wedge_\Sigma \tau &= \tau && \text{if } \Sigma \vdash \tau <: \sigma \text{ and } \Sigma \nvdash \sigma \equiv \tau \\
\sigma \mid \sigma' \wedge_\Sigma \tau &= \sigma \wedge_\Sigma \tau \vee_\Sigma \sigma' \wedge_\Sigma \tau \\
\sigma \wedge_\Sigma \tau \mid \tau' &= \sigma \wedge_\Sigma \tau \vee_\Sigma \sigma \wedge_\Sigma \tau' && \text{if } \sigma \text{ not union} \\
\texttt{Callable}[[\vec{\sigma}], \sigma] \wedge_\Sigma \texttt{Callable}[[\vec{\tau}], \tau] &= \texttt{Callable}[[\sigma_1 \vee_\Sigma \tau_1, \dots, \sigma_n \vee_\Sigma \tau_n], \sigma \wedge_\Sigma \tau] \\
\texttt{tuple}[\vec{\sigma}] \wedge_\Sigma \texttt{tuple}[\vec{\tau}] &= \texttt{tuple}[\sigma_1 \wedge_\Sigma \tau_1, \dots, \sigma_n \wedge_\Sigma \tau_n] \\
\sigma \wedge_\Sigma \tau &= \texttt{Never} && \text{otherwise}
\end{aligned}
$$

elem-type(Σ, τ)	*type*

Type of the elements drawn from τ by a generator, mirroring iter (§A.4). A tuple is iterated at the join

of the base types of its components, and a type with no clause cannot be iterated.

$$\begin{aligned}
\text{elem-type}(\Sigma, \texttt{list}[\tau]) &= \tau \\
\text{elem-type}(\Sigma, \texttt{str}) &= \texttt{str} \\
\text{elem-type}(\Sigma, \texttt{dict}[\texttt{str}, \tau]) &= \texttt{str} \\
\text{elem-type}(\Sigma, \texttt{tuple}[\vec{\tau}]) &= \bigvee\nolimits_{\Sigma} \text{base-type}(\tau_1), \ldots, \text{base-type}(\tau_n) \\
\text{elem-type}(\Sigma, \sigma \mid \tau) &= \text{elem-type}(\Sigma, \sigma) \vee_{\Sigma} \text{elem-type}(\Sigma, \tau)
\end{aligned}$$

$\text{shapes}(\Sigma, \tau, H), \text{shapes}(\Sigma, \vec{\tau})$	*ordered set of shapes or of shape sequences*

Shapes of type τ that remain once the heads in H are excluded. A union has no head to exclude, so the members of the union are taken separately, each with the heads of H typed at that member. A dictionary is refined by its keys rather than by a head, so a dictionary type gives the dictionary form with no key bound. Every other type gives the rest form $\tau \setminus H$, except that no shape remains when H exhausts τ: a literal type whose literal is in H, `bool` when both `True` and `False` are in H, `None` when `None` is in H, a class below a class in H, and `Never`. A sequence of types yields the sequences of shapes of its components.

$$\begin{aligned}
\text{shapes}(\Sigma, \sigma \mid \tau, H) &= \text{shapes}(\Sigma, \sigma, H_\sigma) \cup \text{shapes}(\Sigma, \tau, H_\tau) \\
&\qquad \text{where } H_\sigma = \{h \in H \mid \Sigma \vdash h : \sigma\} \text{ and } H_\tau = \{h \in H \mid \Sigma \vdash h : \tau\} \\
\text{shapes}(\Sigma, \texttt{Literal}[\ell], H) &= \emptyset && \text{if } \ell \in H \\
\text{shapes}(\Sigma, \texttt{bool}, H) &= \emptyset && \text{if } \{\texttt{True}, \texttt{False}\} \subseteq H \\
\text{shapes}(\Sigma, \texttt{None}, H) &= \emptyset && \text{if } \texttt{None} \in H \\
\text{shapes}(\Sigma, C, H) &= \emptyset && \text{if } \Sigma \vdash C <: D \text{ for some } D \in H \\
\text{shapes}(\Sigma, \texttt{Never}, H) &= \emptyset \\
\text{shapes}(\Sigma, \texttt{dict}[\texttt{str}, \tau], H) &= \{\emptyset \setminus_\tau H\} \\
\text{shapes}(\Sigma, \tau, H) &= \{\tau \setminus H\} && \text{otherwise} \\
\text{shapes}(\Sigma, \vec{\tau}) &= \text{shapes}(\Sigma, \tau_1, \emptyset) \times \cdots \times \text{shapes}(\Sigma, \tau_n, \emptyset)
\end{aligned}$$

$\text{shape-type}(k)$	*type*

Least type of shape k, with $\Sigma \vdash k : \text{shape-type}(k)$ (Figure 3.12): the class of a constructor, componentwise for a tuple, and the type recorded by the form otherwise.

$$\begin{aligned}
\text{shape-type}(\tau \setminus H) &= \tau \\
\text{shape-type}(C(\vec{k}) \setminus H) &= C \\
\text{shape-type}((\vec{k})) &= \texttt{tuple}[\text{shape-type}(k_1), \ldots, \text{shape-type}(k_n)] \\
\text{shape-type}([\vec{k}]_\tau) &= \texttt{list}[\tau] \\
\text{shape-type}(\beta \setminus_\tau H) &= \texttt{dict}[\texttt{str}, \tau]
\end{aligned}$$

$\text{seq-safe}(\Sigma, \Gamma, p, \tau)$	*Boolean*

Whether p applies each tuple pattern only to tuples and each list pattern only to lists: no tuple pattern

in p is checked against a type with list values, and no list pattern against a type with tuple values. The list types, `Sized` and `object` have list values; the tuple types, `Sized` and `object` have tuple values; a union has the values of its members. The final case gives tt for a pattern at a type that it cannot match, such as a tuple pattern at a tuple type of another arity.

$$\begin{aligned}
\text{seq-safe}(\Sigma, \Gamma, p, \sigma \mid \tau) &= \text{seq-safe}(\Sigma, \Gamma, p, \sigma) \wedge \text{seq-safe}(\Sigma, \Gamma, p, \tau) \\
\text{seq-safe}(\Sigma, \Gamma, p \texttt{ as } x, \tau) &= \text{seq-safe}(\Sigma, \Gamma, p, \tau)
\end{aligned}$$

$$\begin{aligned}
\text{seq-safe}(\Sigma, \Gamma, (p_1, \dots, p_n), \texttt{tuple}[\tau_1, \dots, \tau_n]) &= \textstyle\bigwedge_i \text{seq-safe}(\Sigma, \Gamma, p_i, \tau_i) \\
\text{seq-safe}(\Sigma, \Gamma, (\vec{p}), \texttt{list}[\tau]) &= \mathsf{ff} \\
\text{seq-safe}(\Sigma, \Gamma, (\vec{p}), \texttt{Sized}) &= \mathsf{ff} \\
\text{seq-safe}(\Sigma, \Gamma, (\vec{p}), \texttt{object}) &= \mathsf{ff}
\end{aligned}$$

$$\begin{aligned}
\text{seq-safe}(\Sigma, \Gamma, [\vec{p}], \texttt{list}[\tau]) &= \textstyle\bigwedge_i \text{seq-safe}(\Sigma, \Gamma, p_i, \tau) \\
\text{seq-safe}(\Sigma, \Gamma, [\vec{p}], \texttt{tuple}[\vec{\tau}]) &= \mathsf{ff} \\
\text{seq-safe}(\Sigma, \Gamma, [\vec{p}], \texttt{Sized}) &= \mathsf{ff} \\
\text{seq-safe}(\Sigma, \Gamma, [\vec{p}], \texttt{object}) &= \mathsf{ff}
\end{aligned}$$

$$\begin{aligned}
\text{seq-safe}(\Sigma, \Gamma, \{\vec{w} : \vec{p}\}, \texttt{dict}[\texttt{str}, \tau]) &= \textstyle\bigwedge_i \text{seq-safe}(\Sigma, \Gamma, p_i, \tau) \\
\text{seq-safe}(\Sigma, \Gamma, q(a), \tau) &= \textstyle\bigwedge_i \text{seq-safe}(\Sigma, \Gamma, p_i, \sigma_i) \\
&\qquad \text{where } \Gamma \vdash q : \mathsf{Type}(C),\ \text{fields}(\Sigma, C) = \vec{x} : \vec{\sigma} \\
&\qquad \text{and field-map}(\Sigma, C, a) = \{\vec{x} : \vec{p}\} \\
\text{seq-safe}(\Sigma, \Gamma, p, \tau) &= \mathsf{tt} \qquad \text{otherwise}
\end{aligned}$$

A.3 Classes and modules

The functions on module names, field lists and classes introduced in §2.4, and the resolution of qualified names in an environment.

submods(q) — *context*

Context that binds each submodule of q to a stub: a context entry $x : \mathsf{Mod}(q.x)$ for each module $q.x$ of the program or of the predefined modules.

$$\text{submods}(q) = \{x : \mathsf{Mod}(q.x) \mid q.x \in \text{dom}(\mathcal{M}) \cup \text{dom}(\mathcal{P})\}$$

own-fields(ϕ) — *sequence of bindings*

Fields declared by a field list, with their types, in order.

$$\begin{aligned}
\text{own-fields}(x_1 : \psi_1 \ \dots \ x_{n+1} : \psi_{n+1}) &= x_1 : \psi_1, \dots, x_{n+1} : \psi_{n+1} \\
\text{own-fields}(\texttt{pass}) &= \varepsilon
\end{aligned}$$

fields(Σ, C) *sequence of bindings*

Fields of class C with their declared types: the fields of its base class, if any, followed by the fields declared by C. Field names are distinct, so the sequence is read as a map by rules that use only the names or only the types.

$$\begin{aligned} \text{fields}(\Sigma, C) &= \vec{x} : \vec{\tau} && \text{if } \Sigma(C) = (\vec{x} : \vec{\tau};\ \bot) \\ \text{fields}(\Sigma, C) &= \text{fields}(\Sigma, D) +\!\!+ \vec{x} : \vec{\tau} && \text{if } \Sigma(C) = (\vec{x} : \vec{\tau};\ D) \end{aligned}$$

ancestors(Σ, C) *set of classes*

Class C together with all its base classes.

$$\begin{aligned} \text{ancestors}(\Sigma, C) &= \{C\} && \text{if } \Sigma(C) = (\vec{x} : \vec{\tau};\ \bot) \\ \text{ancestors}(\Sigma, C) &= \{C\} \cup \text{ancestors}(\Sigma, D) && \text{if } \Sigma(C) = (\vec{x} : \vec{\tau};\ D) \end{aligned}$$

field-map$(\Sigma, C, \vec{u})$ *map from variables*

Argument supplied for each field of C: positional arguments fill the fields in order and keyword arguments match by name. It is undefined unless the arguments cover the fields exactly, so constructor patterns and expressions check and evaluate field by field without regard to the split of the arguments between positional and keyword.

$$\text{field-map}(\Sigma, C, (u_1, \ldots, u_n, y_1{=}u_{n+1}, \ldots, y_m{=}u_{n+m})) = \{x_1 : u_1, \ldots, x_n : u_n, y_1 : u_{n+1}, \ldots, y_m : u_{n+m}\}$$
$$\text{if } \text{dom}(\text{fields}(\Sigma, C)) = x_1 \cdot \ldots \cdot x_n \cdot \vec{y}\,' \text{ and } \{\vec{y}\} = \{\vec{y}\,'\}$$

resolve(ρ, q) *value*

Value that qualified name q denotes in ρ, mirroring static judgement $\Gamma \vdash q : \theta$ (Figure 3.1): a simple name is looked up in ρ directly, and a qualified name in the environment of the loaded module that its prefix denotes. Undefined if the prefix denotes a module stub or a value other than a module.

$$\begin{aligned} \text{resolve}(\rho, x) &= \rho(x) \\ \text{resolve}(\rho, q.x) &= \rho'(x) && \text{where } \text{resolve}(\rho, q) = \mathsf{Mod}(q', \rho') \end{aligned}$$

name(q) *string*

String literal naming module q, so that name(`__main__`) = `"__main__"`.

A.4 Values

The functions on values used by the evaluation rules. The comparisons `==`, `!=`, `in` and `not in` are given in terms of structural equality eq and membership contains. When defined, the operators agree with Python, yielding $\mathsf{aborts}\ \kappa$ when Python raises an exception. When undefined, evaluation of the expression has no derivation, and an implementation may abort or produce the result that Python gives (Chapter 1). Because equality is partial, the order in which elements are compared is observable:

equality and membership short-circuit at the first element that decides the result, so an undefined comparison later in a sequence does not make the whole comparison undefined.

$\hat{\odot}(v, v')$	*result*

Meaning of a strict binary operator.

$\odot$	$\hat{\odot}(v, v')$
`==`	val `False` if both operands are `nan`, and val $\text{eq}(v, v')$ otherwise
`!=`	val `True` if both operands are `nan` or $\text{eq}(v, v') = \texttt{False}$, and val `False` if $\text{eq}(v, v') = \texttt{True}$
`in`	val $\text{contains}(v', v)$
`not in`	val `True` if $\text{contains}(v', v) = \texttt{False}$, and val `False` if $\text{contains}(v', v) = \texttt{True}$
`<` `<=` `>` `>=`	[**TODO:** Ordering, as in Python]
`+` `-` `*` `/` `//` `%` `**`	[**TODO:** Arithmetic, as in Python. Float precision is left to the implementation. aborts ZeroDivisionError where the divisor is zero, and aborts TypeError where an operand is not a number]

$\hat{\ominus}(v)$	*result*

Meaning of a unary operator.

$\ominus$	$\hat{\ominus}(v)$
`not`	val `True` if $v = \texttt{False}$, and val `False` if $v = \texttt{True}$
`+`	val v if v is a number, and aborts TypeError otherwise
`-`	val $-v$ if v is a number, and aborts TypeError otherwise

$\text{eq}(v, v')$	*Boolean*

Structural equality: literals by value, lists and tuples by their elements, dictionaries with the same keys by their values, and objects of the same class by their field values, each through eq-elems. `None` compares with any value; otherwise equality is undefined between values of different kinds. Dictionaries are compared in the entry order of the left operand.

$$
\begin{aligned}
\text{eq}(\ell, \ell) &= \texttt{True} && \text{if } \ell \text{ is not } \texttt{nan} \\
\text{eq}(n, n') &= \texttt{False} && \text{if } n \neq n' \\
\text{eq}(w, w') &= \texttt{False} && \text{if } w \neq w' \\
\text{eq}(\texttt{None}, v) &= \texttt{False} && \text{if } v \neq \texttt{None} \\
\text{eq}(v, \texttt{None}) &= \texttt{False} && \text{if } v \neq \texttt{None} \\
\text{eq}(\texttt{True}, \texttt{False}) &= \texttt{False} \\
\text{eq}(\texttt{False}, \texttt{True}) &= \texttt{False}
\end{aligned}
$$

$$
\begin{aligned}
\text{eq}([\vec{v}], [\vec{v}\,']) &= \text{eq-elems}(\vec{v}, \vec{v}\,') \\
\text{eq}((\vec{v}), (\vec{v}\,')) &= \text{eq-elems}(\vec{v}, \vec{v}\,')
\end{aligned}
$$

$$\begin{aligned}
\text{eq}(\delta, \delta') &= \text{eq-elems}(v_1 \cdot \ldots \cdot v_n, \delta'(w_1) \cdot \ldots \cdot \delta'(w_n)) && \text{if } \text{dom}(\delta) = \text{dom}(\delta') \\
&\quad \text{where } \delta = w_1 : v_1 \cdot \ldots \cdot w_n : v_n \\
\text{eq}(\delta, \delta') &= \texttt{False} && \text{if } \text{dom}(\delta) \neq \text{dom}(\delta')
\end{aligned}$$

$$\begin{aligned}
\text{eq}(\mathsf{Obj}(C, \rho), \mathsf{Obj}(C, \rho')) &= \text{eq-elems}(\rho(x_1) \cdot \ldots \cdot \rho(x_n), \rho'(x_1) \cdot \ldots \cdot \rho'(x_n)) \\
&\quad \text{if } \text{dom}(\text{fields}(\Sigma, C)) = x_1, \ldots, x_n \\
\text{eq}(\mathsf{Obj}(C, \rho), \mathsf{Obj}(D, \rho')) &= \texttt{False} && \text{if } C \neq D
\end{aligned}$$

eq-elems($\vec{v}, \vec{v}\,'$) — *Boolean*

Equality of two sequences of values, compared from the first element: `False` at the first unequal pair or if the lengths differ, and `True` when every pair is equal.

$$\begin{aligned}
\text{eq-elems}(\varepsilon, \varepsilon) &= \texttt{True} \\
\text{eq-elems}(v \cdot \vec{v}, v' \cdot \vec{v}\,') &= \texttt{False} && \text{if } \text{eq}(v, v') = \texttt{False} \\
\text{eq-elems}(v \cdot \vec{v}, v' \cdot \vec{v}\,') &= \text{eq-elems}(\vec{v}, \vec{v}\,') && \text{if } \text{eq}(v, v') = \texttt{True} \\
\text{eq-elems}(\vec{v}, \vec{v}\,') &= \texttt{False} && \text{if } |\vec{v}| \neq |\vec{v}\,'|
\end{aligned}$$

contains(v, v') — *Boolean*

Whether v' is an element of v: of a list or tuple by structural equality, of a dictionary as a key, and of a string as a substring.

$$\begin{aligned}
\text{contains}([\vec{v}], v) &= \text{contains-elems}(\vec{v}, v) \\
\text{contains}((\vec{v}), v) &= \text{contains-elems}(\vec{v}, v)
\end{aligned}$$

$$\begin{aligned}
\text{contains}(\delta, w) &= \texttt{True} && \text{if } w \in \text{dom}(\delta) \\
\text{contains}(\delta, w) &= \texttt{False} && \text{if } w \notin \text{dom}(\delta)
\end{aligned}$$

$$\begin{aligned}
\text{contains}(w', w) &= \texttt{True} && \text{if } w' = w_1 \mathbin{+\!\!+} w \mathbin{+\!\!+} w_2 \text{ for some } w_1, w_2 \\
\text{contains}(w', w) &= \texttt{False} && \text{otherwise}
\end{aligned}$$

contains-elems($\vec{v}, v$) — *Boolean*

Whether a sequence of values has an element structurally equal to v, taking the elements in order and stopping at the first element equal to v.

$$\begin{aligned}
\text{contains-elems}(\varepsilon, v) &= \texttt{False} \\
\text{contains-elems}(v' \cdot \vec{v}, v) &= \texttt{True} && \text{if } \text{eq}(v, v') = \texttt{True} \\
\text{contains-elems}(v' \cdot \vec{v}, v) &= \text{contains-elems}(\vec{v}, v) && \text{if } \text{eq}(v, v') = \texttt{False}
\end{aligned}$$

elems(v) *sequence of values*

Elements of a list or tuple, or the characters of a string, in order.

$$\begin{aligned}
\text{elems}([\vec{v}]) &= \vec{v} \\
\text{elems}((\vec{v})) &= \vec{v} \\
\text{elems}(w) &= w_1 \cdot \ldots \cdot w_m \qquad \text{where } w = w_1 \mathbin{+\!\!+} \ldots \mathbin{+\!\!+} w_m, \text{ each } w_i \text{ one character}
\end{aligned}$$

iter(v) *sequence of values*

Sequence drawn from v by a generator: the keys when v is a dictionary, and otherwise the elements of v.

$$\begin{aligned}
\text{iter}(\delta) &= w_1 \cdot \ldots \cdot w_n \qquad \text{where } \delta = w_1 : v_1 \cdot \ldots \cdot w_n : v_n \\
\text{iter}(v) &= \text{elems}(v) \qquad \text{otherwise}
\end{aligned}$$

getitem(v, v') *result*

Subscripting: the value under key v' of a dictionary, or the element at index v' of a sequence, counting from the end when the index is negative. Aborts with KeyError, IndexError or TypeError when Python raises the exception of that name.

$$\begin{aligned}
\text{getitem}(\delta, w) &= \mathsf{val}\ \delta(w) && \text{if } w \in \text{dom}(\delta) \\
\text{getitem}(\delta, w) &= \mathsf{aborts\ KeyError} && \text{if } w \notin \text{dom}(\delta) \\
\text{getitem}(v, n) &= \mathsf{val}\ v_{n+1} && \text{if elems}(v) = v_1 \cdot \ldots \cdot v_m \text{ and } 0 \le n < m \\
\text{getitem}(v, n) &= \mathsf{val}\ v_{m+n+1} && \text{if elems}(v) = v_1 \cdot \ldots \cdot v_m \text{ and } -m \le n < 0 \\
\text{getitem}(v, n) &= \mathsf{aborts\ IndexError} && \text{if elems}(v) = v_1 \cdot \ldots \cdot v_m \text{ and } (n < -m \text{ or } n \ge m) \\
\text{getitem}(v, v') &= \mathsf{aborts\ TypeError} && \text{otherwise; } v \text{ not a dictionary}
\end{aligned}$$

result(o) *result*

Result of a call from the outcome of its body: the value returned, or `None` if the body completes without returning, or the abort if the body aborts.

$$\begin{aligned}
\text{result}(\mathsf{returns}\ v) &= \mathsf{val}\ v \\
\text{result}(\mathsf{assigns}(\rho)) &= \mathsf{val}\ \texttt{None} \\
\text{result}(\mathsf{aborts}\ \kappa) &= \mathsf{aborts}\ \kappa
\end{aligned}$$

dispatch($\rho, v, \vec{p}, \vec{s}$) *bindings and statement*

Case taken by a `match` statement for value v: the bindings and statement of the first pattern that

matches, or no bindings and `pass` if no pattern matches.

$$\begin{aligned}
\text{dispatch}(\rho, v, \varepsilon, \varepsilon) &= (\emptyset, \texttt{pass}) && \\
\text{dispatch}(\rho, v, p \cdot \vec{p}, s \cdot \vec{s}) &= (\rho', s) && \text{if } \rho, p, v \rightsquigarrow \textsf{match } \rho' \\
\text{dispatch}(\rho, v, p \cdot \vec{p}, s \cdot \vec{s}) &= \text{dispatch}(\rho, v, \vec{p}, \vec{s}) && \text{if } \rho, p, v \rightsquigarrow \textsf{no-match}
\end{aligned}$$

update(δ, δ') *dictionary entries*

Dictionary entries δ updated by the bindings of δ' in order, each replacing any earlier entry under the same key.

$$\begin{aligned}
\text{update}(\delta, \varepsilon) &= \delta \\
\text{update}(\delta, w : v \cdot \delta') &= \text{update}(\text{setitem}(\delta, w, v), \delta')
\end{aligned}$$

setitem(δ, w, v) *dictionary entries*

Dictionary entries δ with w bound to v, replacing any earlier binding of w.

$$\begin{aligned}
\text{setitem}(\varepsilon, w, v) &= w : v && \\
\text{setitem}(w : u \cdot \delta, w, v) &= w : v \cdot \delta && \\
\text{setitem}(w' : u \cdot \delta, w, v) &= w' : u \cdot \text{setitem}(\delta, w, v) && \text{if } w \neq w'
\end{aligned}$$

Annex B

Excluded Python behaviours (informative)

This annex is informative and imposes no requirements. It summarises the Python behaviours excluded by PurePy and the stage at which each exclusion takes effect; the normative basis is the well-formedness rules (Chapter 3) and the operational semantics (Chapter 4).

B.1 Mutable state

Figure B.1 lists the Python constructs associated with mutation and shared state, and says how and when PurePy excludes each.

Construct	Python	Exclusion	Mechanism
`global` and `nonlocal`	Direct assignments to variable of enclosing scope	Syntactic	Not in grammar
`del`	Removes binding	Syntactic	Not in grammar
Augmented assignment `+=`, etc.	Updates in place if type supports it	Syntactic	Not in grammar
Walrus operator `:=`	Assigns within expression	Syntactic	Not in grammar
Item assignment `xs[0] = v`	Updates container in place	Syntactic	Not in grammar; assignment targets are variables
Attribute assignment `p.x = v`	Updates object in place	Syntactic	Not in grammar; assignment targets are variables
Mutating methods, such as `list.append`	Update value in place	Dynamic	Attribute reference undefined
Reassignment of captured variable	Closure sees new value	Static	Variable captured by statement may not be assigned later in sequence, nor by statement itself
Lambdas capturing comprehension variable	Variable mutated on each iteration; lambda sees final value	Static	Lambda may not capture generator variable

Figure B.1: Python mutation of state excluded by PurePy

B.2 Implicit coercions

Python accepts several implicit coercions and lax matches. Figure B.2 lists those excluded by PurePy, and records at which stage each exclusion takes effect. A dynamically excluded operation leaves the run without a result, so the program runs under Python but not under PurePy.

Construct	Python	Exclusion	Mechanism
Tests of `if` and `assert`	Coerced to `bool`	Dynamic	Must already be Boolean
Sequence patterns	List patterns match tuples, and tuple patterns lists	Dynamic	List patterns match only lists, tuple patterns only tuples
Constructor patterns	Positional class patterns may supply fewer sub-patterns than class has fields	Static	Require exact arity
Dictionary keys	Any hashable value, with `1`, `1.0` and `True` equivalent as keys	Static in patterns, dynamic in expressions	Strings only
`assert` message	Any object, shown in traceback	Dynamic	Strings only
Duplicate name in `from` import	Repeated name rebinds harmlessly	Static	Imported names distinct
`==` and `!=`	Compare any two operands, giving `False` when unrelated	Dynamic	Undefined between, for example, number and string, or list and tuple; only `None` compares with anything
Functions under `==`	Compared by identity, so two textually identical lambdas are unequal	Dynamic	Equality undefined
`nan` comparison via container equality	Elements compared by identity before equality, so identical NaNs match but distinct ones do not	Dynamic	Undefined when both elements are `nan`

Figure B.2: Python coercions and lax matches excluded by PurePy

B.3 Runtime errors

A well-formed PurePy program avoids the Python runtime errors listed in Figure B.3, under the conditions noted there. Well-formedness does not prevent errors from ill-typed operations, such as wrong operand types, calling a non-function, calling a function with the wrong number of arguments, or attribute access on an object lacking the attribute. Each of these has a termination kind and a rule in the semantics (Figure 4.10).

Python error	Ruled out when	See
`NameError`, `UnboundLocalError`	Always: a variable may be referenced only where it is definitely assigned.	Figure 3.9, Figure 3.4
`ModuleNotFoundError`	Always: an import requires its target to be a module of the program.	Figure 3.16
`ImportError`	Always: a from-import requires each imported name to be a member of the source module, and import cycles are rejected. (A Python cycle may run or fail, depending on where the cycle is entered.)	Figure 3.15, Figure 3.17
`AttributeError`	The reference is a module attribute, which must resolve to a definitely assigned member.	Figure 3.4
`TypeError`	The operation is a constructor call or class pattern, which must match the arity of the class exactly. (Only excess sub-patterns are a Python error; a pattern with fewer sub-patterns is excluded even though Python permits it.)	Figure 3.4, Figure 3.13

Figure B.3: Python errors prevented by well-formedness

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